

**Factors Affecting Mode Choice of Work Trip Using
Multinomial Logit Model: A Case Study of Agrabad,
Chittagong.**

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ABSTRACT

The industrial sector plays a crucial role in economic growth of Bangladesh. It is the highest contributor in generating currency & employment. Chittagong, the commercial capital of Bangladesh, has a diverse range of industries, is chosen as study area. Employed individuals make various working and non-working trips using various modes of transportation. This study aims to analyze the factors affecting the mode choice and determine the travel satisfaction level of employee people. A sample data of 550 workers are collected randomly through a questionnaire survey in Agrabad, Chittagong. The multinomial logistic model is applied for analyzing mode choice and conjoint analysts & weighted index model is applied for determining significant travel satisfaction factor and satisfaction level. The outcomes showed that majority of them use bus for working trip and walking for non-working trip as travel mode. Considering bus and leguna. as reference, factors are affecting the mode choice of the employed people for working trips are gender, income level, length of trip, direct trip time & fare and for non-working top excluding gender others factors are same as working trips. The utility of the travel satisfaction is getting higher as it is moving from poor to excellent for accessibility, safety, security & reliability, high to less for cost and uncomfortable to comfortable. Safety & security are the most and accessibility is less significant satisfactory factor for bus. Among the factors, workers are mostly satisfied with the reliability of the modes and satisfactory score for work and non-work trip mode is 3.28 and 3.41 respectively which indicates that workers are very satisfied with both of the modes. The study will be helpful for developing the existing transportation facilities and amenities which conditions are inferior and have negative impact on mode choice and travel satisfaction of employed people.

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Chapter -1

Introduction

1.1 Background

Transport is one of the most important aspects in the day-to-day life of the people, as their activities cannot be fulfilled without moving (MADHUWANTHI et al., 2016). The availability of the travel modes is different and people can choose the mode with their desire to complete their trip. The traveling modes are now becoming different aspects from the conventional models. Now peoples tend to choose the mode concerning comfort, security, vehicle time, trip distance, time reliability, cost of the travel mode etc. (MADHUWANTHI et al., 2016a).

The transportation system has an imperative role in the growth and development of a country. It is one of the most important parts of the whole urban system. The role of the transportation system is to provide links between different urban sectors. It is very important to have an efficient urban transport system to facilitate access to different sectors. The development of an urban area will sustain only if it can provide efficient and effective access to all its parts. An efficient and effective urban transport system means reaching any destination point from any origin within the urban boundary with minimum travel time and delay time. Proper transportation planning can provide efficient movement of both people and goods as well as can ensure access to all sectors of an urban area.

Bangladesh has started developing a vision for implementing sustainability in its transport sector, although it is one of the world's most densely populated countries(Mohiuddin et al., 2022). Every year Bangladesh faces high crash and mortality rates, the high level of motorized traffic congestion and related air pollution, and deteriorating public health conditions(Mohiuddin et al., 2022).Chittagong is well-known as the country's commercial hub and a major contributor to national exports and imports, with strong ties to the home economy(Sifat et al., 2021). Mode choice behaviors of citizens are quite different and it is necessary to define the causes behind this particular choice in their travel mode(Sifat et al., 2021).Income, household type and travel distance play a significant role in transport mode choice. People trend to buy cars and alter their transport mode choice behavior while their income is being increased(Almasri & Alraee, 2013). Agrabad is a downtown commercial and financial area in Chittagong, the second largest city of Bangladesh. Located close to the city's harbor, it hosts numerous Bangladeshi and international businesses, banks,

financial institutions, and the other commercial entities. The Chittagong Chamber of Commerce & Industry, the World Trade Center and Chittagong Stock Exchange are most prominent of them. The south & north Agrabad ward consist of 175,268 people(South Agrabad, n.d.; Wikipedia North Agrabad, n.d.).There is a rush in the employed class as they have to use different vehicle to reach their destination. But as there are different type of motorized vehicle in Chittagong city like bus, rickshaw, CNG and also non-motorized mode or active transportation modes like walking, cycling can be used for reaching to destination, it can be a confusing situation of the workers to choose their modes of travel. Now a day, due to urbanization number of public and private modes are increasing day by day. These increasing number of modes are causing traffic congestion which make people late to reach to their destination. In this point of view non-motorized vehicle can be useful. But to reach a long distance non-motorized vehicle shall not be useful. Rather than this, economic status, age, vehicle ownership and many other factors can effect on the mode choice of the workers. On the other hand, satisfaction of the passengers of different modes is also an important factor that effect the travel mode choice. Thus, for understanding and finding out the key factors for which people choose different mode, for finding out the mostly used mode of the workers for further development of that mode and to know about their travel satisfaction it is necessary to study about the factors affecting mode choice and travel satisfaction.

1.2 Research Goal

- To examine the factors that influence the mode choice of employed people in work trips of Agrabad in Chittagong City.

1.3 Research Objective

- To analyze the significant factors that affect travel mode choice behavior using multiple linear regression.
- To determine the travel satisfaction factors & the satisfaction level in Agrabad commercial area in Chittagong using conjoint analysis.

1.4 Rationality

- Mitigating the transportation problems resulting from urbanization and rapid increase of population.
- Predicting the mode choice behavior of employed people in the Agrabad area.
- Travel demand analysis & future modal split can be predicted through this study.
- Mode choice modeling is very important for the future growth of each mode & it is very important to understand the travelers' needs and their preferences of using the modes.
- Specifying the factors that contribute to the use of each mode and shifting from one mode to another.

1.5 Scope of the study

In the Chittagong city, the number of modes are being increased day by day. This is causing traffic jams and making the workers late for their work. Some of the modes are not really useful and not used by most of the workers. Besides, employed people do not feel reliable, comfortable to travel with every mode. To understand their mode choosing pattern and finding out the mostly used mode by the workers, some important factors are need to be identified by which choice of mode and satisfaction are mainly affected. The scopes of the study are explained as follows:

- This study will provide a comprehensive outlook of the factor affecting the mode choice and mostly used mode and the findings will be useful for further research on working on that those modes particularly and managing as well as developing one or two particular modes for the trip of the garment worker people.
- Besides this study will also find out the most important satisfactory factor among comfortability, reliability, safety, security, accessibility. Thus, the findings will be useful for further research on that factor.
- The overall satisfaction of the worker will be determined. Trip pattern of the worker people and number of trips of the worker in a day will be provided from this study. These findings can be used to examine the traffic jam because of the employed persons' trip in further research.
- The mode choice of both working and non-working trips shall be determined.

- The study may help in developing the land use pattern in future which is needed for balancing the mixed land use the output of the study may be used for planning the road network system.
- This type of study will be helpful to understand the importance of both motorized and non-motorized transportation and their user's satisfaction level of these modes in order to contribute to sustainable transportation development of that area.

1.6 Limitation of the study

Though this study is very important for identifying the travel demand and behavior of the people, it has some limited boundary due to lack of efficient secondary data and other criteria. The limitations are described in the below:

- Build-environment and self-selection factors (e.g., attitude) are not considered in the study.
- Required a large number of data and these secondary data are not available at the all-time.
- Due to time duration and lack of manpower, this study cannot be done on all the employed peoples mode choice of Chittagong city.
- This work is done only for a part of the Chittagong city and considered people for information is relevant to banks & other financial institutions only.
- The respondents are not co-operative sometimes and felt hesitation in giving their personal information.
- The actual route used by the worker to reach to their destination is difficult to identify. That's why a hypothesis is taken that workers use shortest route which is needed to identify their trip pattern.

Chapter -2

Literature Review

2.1 Transportation

Transport is as the department of movables and individuals from one space to another and the several means that helps in accomplishment of such department. The term is derivative from the Latin trans ("across") and portare ("to carry") (World Bank, 2008). The transportation system is defined as the displacement of goods or passengers from one place to another. It has always played a significant role in a country's economic, social and financial growth. Even in today's competitive market scenario, many situations came across where the decisions need to be taken to give the best solutions for the problem out of all possible solutions. Therefore, it is essential for an effective transportation system and its smooth functioning to have a good transportation model(Kacher & Singh, 2021). It is also demarcated as part of the economic movement anticipated to gratify social essentials through changing of persons or belongings positions geographically. For growth of any nation, well-organized and efficient transport management system is an obligatory as an elementary infrastructure for development. Throughout the antiquity, it has been perceived that for any nation progress a suitable, widespread and effectual road transport has played a vital part. Transporters at every stage of advanced civilization, carry out one of the most significant events. Where streets are deliberated as veins and ribs of a nation, commuters and belongings transported are linked to blood in movement. Passenger Road Transport Service (PKTS), which is associated to the fiscal expansion of a nation, is a crucial. With the help of transportation, persons not just associate but progress. Public's progression has been continual on the expediency, rapidity and security of the means of transport (World Bank, 2008). Transportation of belongings and persons is very imperative in corporate. As raw supplies arrive the location of production, completed merchandises reach the location of transaction or consumption, folks travel from place to place to operate the trade etc.(Siedler, 2014).The development of the capacity and the requirement to carriage enormous masses of things or numbers of individuals over elongated spaces at high rapidity in case and care has been a key of advancement and in certain of technological advancement.

2.2 Mode of Transportation

Transport in Bangladesh is an important part of the nation's economy. Since the liberation of the country, the development of infrastructure within the country has progressed at a rapid pace, and today there is a wide variety of modes of transport by road, railway, air and water are widely used in carrying both passengers and cargo (Islam & Hoque, n.d.-a).

2.3 Road Transport

Road Transport Road transport in Bangladesh is a private sector affair operating predominantly in domestic routes. Rates are among the cheapest in the world. Express and nonstop services are available to principal towns from Gabtoli, Saidabad and Mohakhali bus terminals in Dhaka. The Bangladesh Road Transport Corporation (BRTC) also maintains a countrywide network of bus services. Recently they have introduced Dhaka - Calcutta - Dhaka direct daily bus services via Benapole, Jessore (Islam & Hoque, n.d.-b).

2.4 Railway Transport

Railway acts as an important method of mass transport in Bangladesh. Many districts of the country are connected via railways. Bangladesh Railway was mostly inherited from the British-established Assam Bengal Railway system after the partition of India in 1947. Bangladesh Railway's headquarters are located in the southern port city of Chattogram. About 32% of the total area of Bangladesh is effectively covered by the railways. As of 2005, the total length of railway is 2,706 kilometers (1,681 mi). Of that, 923 km (574 mi) are broad gauge (1,676 mm) tracks (mostly in the western region), and the remaining 1,822 km (1,132 mi) are meter gauge tracks (mostly in the central and eastern regions). The Bangladesh Railway provides an efficient service to places of interest such as Chattogram, Sylhet, Khulna, Mymensingh, Bogura, Rajshahi, Dinajpur starting from Dhaka. The inter-city Express Service is available to and from important cities at cheap fares (Islam & Hoque, n.d.-b).

2.5 Air Transportation

Air transport is mainly characterized by safety, high speed and high level of customer service, but its availability is limited. Air transport is also considerably expensive. For these reasons, air transport is particularly suitable for transport of objects of high values for long distances. Biman Bangladesh Airlines is the national airlines of Bangladesh which commenced operation in 1972. There are currently three international airports in

Bangladesh: Shahjalal International Airport in Dhaka, Shah Amanat International Airport in Chattogram and Osmani International Airport in Sylhet. All three international airports have direct connections to a number of destinations in the Middle East while Hazrat Shahjalal International Airport has services to the wider Asian region and Europe. In addition to the international airports there are also five domestic airports in Barisal, Cox's Bazar, Jessore, Rajshahi and Saidpur with almost all services from these airports either bound for or originating from Dhaka (Islam & Hoque, n.d.-b).

2.6 Trip

According to National Household Travel Survey (2001), movement of an individual from one location to another is defined as trip (Mcguckin et al., n.d.). Also, trip is defined as an uninterrupted series of movement stages between two activities. Direct trip is defined as a movement that occurred directly between two anchor destinations, such as a movement from household to work place and a primary or extensive trip destination is known as anchor. A succession of short trips which are connected together among anchor destinations, such a series of trips in a home to work trip such as a trip that leaves household, halts to drop a commuter, halts for drinking coffee and carry on to work (Mcguckin et al., n.d.).

2.7 Mobility

The verbal meaning of mobility is the quality or condition of being mobile" and mobile literal meaning is being capable of moving or being moved willingly from location to another. In the Oxford English Dictionary "mobility is defined as the capability to move or to be moved (Handy, 2002). In the perspective of transport planning, the potential for movement, the capability of movement from one location to another for any purpose is defined as mobility (Handy et al., 1997). The measurement of mobility in transportation planning are traditional methods of level-of-service (LOS). Greater volume-capacity fictions indicate slow-footed mobile times, less easiness of movement, which results in lower mobility rate. Actual movement also used for measuring mobility Either numbers of total trips generated or total distance traveled measures it. But it is not essential that actual movement measurement is accurate measurement for measuring the potentiality of movement (Handy et al., 1997)

2.8 Mode Choice

Mode choice is the process where the means of traveling is determined (Singleton, 2000). The means of travel is devoted to the travel mode, which may be by private automobile, public transportation, walking, bicycling, or other means. The selection of transport means is possibly one of the utmost significant classic methods in transportation planning. It is due to the crucial part portrayed by public transportation in policy designing means make more efficient use of roadways than private transportation. Also, public transportation have more societal reimbursements like if other people commence to his public transportation, congestion rate will be a smaller amount on the highways and the mishaps will be reduced. With public transportation low cost travelling and more efficient fuel use is possible. Key features of public transportation are that they will maintain some specific timetable, regularity etc. Alternatively, private transportation is extremely lithe. It delivers more comfort and fitting travel and better approachability also. The concern of means choice is possibly the only furthestmost significant component in transportation development and policymaking. It disturbs the overall effectiveness in urban regions. It is imperative to develop and use models that are penetrating to individuals travel characteristics that effect distinct selections of mode (Introduction to Transportation Engineering, 2007a).

2.9 Mode Choice in Bangladesh

In Bangladesh, some demand models of travel have been established for efficient transport management and better mode choice for the citizen (Enam & Choudhury, 2011). Four-step travel demand modeling which is one of the well-known travel demand model was adopted for study of Integrated Transport for Metropolitans Area of Dhaka city at earlier which abridged modal choice into binomial choice modelling which divided mode choice between public and private means of transport (Enam & Choudhury, 2011). Another study of modal choice, which is undertaken for Dhaka city, also established four step travel demand model. The study outcomes expressed that factors such as travel time and cost has positive relationship with mode choice, Comfort factor of mode choice has higher values than modes travel time and travel cost. Moreover, according to the research for developing country like Bangladesh this outcome is not normal (Mohammed & Habib, 2002). A study of mode choice for educational trip for Khulna city investigate existing travel pattern of

students from different educational institute and parameters affecting their mode choice. Analyzing the survey data, they found that different socio-economic factor such as age, gender, income level, earning person in the household influence the modal choice of education trip but the study does not recognized self-selection and built-environment parameters (Tahsin,n.d.).In Strategic Transport Plan (STP) for Dhaka, for the most all-encompassing travel demand model a greater domestic survey was directed in which World Bank contribute financially In the research two modes were considered for modal choice model which are public transit and personalized motorized modes. In the personalized motorized mode group, grouping of private cars and taxis were done together overlooking their dissimilar characteristics such as travel cost, obtainability, accessibility etc non-motorized modes are excluded from the mode choice model even though non-motorized vehicle produce 37%-person trip in the total trip generated. The study of mode choice was developed using binominal logit model, which consider two explanatory factors such as travel time and cost for three different income group(THE STRATEGIC TRANSPORT PLAN FOR DHAKA, n.d.). A rule-based mode choice model adopted which only consider cars users assuming that they won't use any other means of transport for their travel except car and a multinomial logit (MNL) model adopted for the other mode choices which are rickshaw, auto-rickshaw, taxi, and bus considering that these four modes are available to all commuters, for development of a four-step travel demand model for mode choice modelling. The model used STP data only by upgrading level-of-service, two parameters travel time, and travel cost were only used for specification of the model (Hasan, 2007).

2.10 Travel Satisfaction

Satisfaction is a decision that a goods or amenity offers a congenial level of consumption associated achievement. Travel satisfaction is also a decision which is offered by different socio-economic or transportation goods and services to a pleasurable level of consumption during a trip (Ettema et al., 2010).As transportation is an important elements in furnishing functional city, it indicates the significance of improving transportation sectors and improving the travel satisfaction of the people of city(Cao & Ettema, 2014).In many studies it is found that satisfaction with travel is in some extent related with transportation infrastructure development (Cao & Ettema, 2014; Olsson et al., 2013).Satisfaction with travel might be inclined by features of the transportation arrangement and accessibility (Cao & Ettema, 2014).In a study of assessing travel satisfaction, consumer based on eminence, safety & security and service delivered by existing bus service in Dhaka city

indicated that majority of the Survey respondent thought that the existing bus service condition is disappointing. In the aspect of safety & security of existing bus service, the core three causes of disappointment of commuters are dangerous driving practices, underprivileged boarding and settling amenities and deficiency of law imposing agencies observation. The provision of service insufficient and unsatisfactory because of poor condition of body and sitting organization of the bus services and discomfort during travel. But user has an optimistic outlook for those bus services which has low travel cost (Rahman et al., 2017). In a study travel satisfaction assessed using leisure trips from Ghent (Belgium) data Different travel mode like car, public transit, walking etc. difference in reliability and structure of satisfaction with travel scale also are considered in the assessment. The outcomes suggest that single aspect specification fit the data than two aspects for all modes of transport as well as separately for car and cycling. It is found that a tri-dimensional configuration is more applicable for public transit and walking though as expected person's substances do not fit on the two emotional dimensions (de Vos et al., 2015)

2.11 Transport & Mobility

For assisting people's activities transportation plays one of the vital role because without any movement it is impossible one's for fulfilling his needs (*Introduction to Transportation Engineering*, 2007b) (Dewi & Munawar, 2010). The need of transportation is because all the human needs most of the time is not provided in in any place, they are provided in different places (Sholichin, 2011). With economic and social interaction people's needs are closely related. Different social demographic characteristics such as age, gender, marital status, occupation, education level as well as activities performed by family member are closely related with people's movements pattern (Dewi & Munawar, 2010) (Eriksson et al., 2010). In participating or conducting activities in different place is associated with people's social obligations, biological needs and personal desires (Eriksson et al., 2010) (Dewi & Munawar, 2010). Commonly many activities are done with variety of purposes, which are working activity, education activity, recreation activity, and social activity (Dewi & Munawar, 2010; MADHUWANTHI et al., 2016b). The mode choice is probably one of the most important sectors in transportation planning In large cities as a result of mode choice effect on social, economic and geographic development, mobility is becoming progressively vital (Albalade & Bel, n.d.). A mode of transport terms as an elucidation that makes use of a particular type of vehicle, infrastructure and operation. People concern on choice of travel modes are dependent on their purpose that required to be fulfilled. Travel

mode availability is differed with type and each mode has its advantages and disadvantages and people can make selection of their desire mood of travel to complete their trip (MADHUWANTHI et al., 2016b) , (Albalate & Bel, n.d.). Since mode choice affects how efficiently one can travel, how much space of the area is dedicated to transportation functions and facilities as well as the variety of options available to the traveler, understanding mode choice is important (Ortizar & Willumsen, 2011). (Dewi, 2010). According to a study most of the travel modes can make a realistic & and reasonable claim that they will be capable to satisfy all community trip purposes (Dewi, 2010) In different studies it is found that peoples have the tendency to choose their travel modes with concerning their security, comfort, trip-time, trip cost, time reliability etc. (Eriksson L, 2008),(Madhuwanthi et al., 2016) (Minal, 2014). People select trip modes that respond best to situations with perceptible needs. Travel mode choice options are more complicated with respect to the user group as people have tendency in differing expectation in choice of trip mode (Dewi, 2010), (Grave, 2003). In studies it is evaluated that for selection of the most suitable trip mode opportunity for every activity is more open and wider from the private transportation to public transportation (Eriksson L, 2008), (Dewi, 2010). In provision of successful transport services decision maker must ensure that service provided by them is more suitable for the consumers.

2.12 Trip-end Modal Split Models

Usually, to estimate the progress in car trips demand so that planning in investment could meet the demand is the major objective of transportation planning. When individual individualities were assumed as the furthestmost significant elements of modal choice, afterwards trip generation efforts were taken to apply modal-split models instantaneously. One of this such a model is entitled as trip-end modal split model. By this approach, different features of the individual could be conserved and used for estimation of modal split the recent models of modal split are related with the selection of mode(Mathew, 2017).

2.13 Trip-interchange Modal Split Models

Interchange provides more flexibility as passengers potentially have more feasible routes to choose from (Correia, 2016). Dividing the "trip interchanges" or Origin-Destination courses calculated by the trip distribution model (Correia, 2016) The main benefit of this model is that include of the features of the trip is possible and that of the substitute means of transport obtainable to commence them. As origin-destination flows are identified, it can calculate travel times, expenditures, etc. efficiently for the contending modes (Correia, 2016). Also, this model is sensitive to trip time, cost etc., it can be possible to use them in assessing the effects of wide range of transport policies such as improvement of transport model, road pricing, transport fare policy, parking related policies etc. (Correia, 2016). This has greater beneficially in long term modeling of mode choice (Mathew, 2017). The number and explanation of means demonstrated depends on the application of problem, obtainable data, system modeling competences, etc. (Correia, 2016). In this model a set of mode of transport is included. In a decision tree of modes choice by using the included modes for a trip in which each node of the decision tree will be represented an alternative mode choice for trip and relationship among the different modes of transport (Correia, 2016).

Chapter -3

Study Area Profile

3.1 Introduction

Each city has its own unique character, which is shaped by the culture, society, climate, vegetation economy, topography, and circumstances of that particular part of the region. There are general description, demographic information, income level, transportation, land-use & road network of the study area.

3.2 General Description

Agrabad is divided into two wards. South Agrabad Ward (27no.ward) is located in the western part of Chittagong City Corporation. To its south is Gosaildanga Ward No. 36, Pathantuli Ward No. 28 to the east, Agrabad Ward No. 24 to the north and North Madhya Haliashahar Ward No. 37 to the west. The coordinates of this ward is 22°19'32"N 91°48'26"E. The area of South Agrabad ward is 1.43 sq km. According to the population census of 2011 it has a population of 66,755. Among them 35,053 males and 31,702 females. Total families are 14,695.

North Agrabad (Ward No. 24) is under Chittagong City Corporation. The administrative activities of the eastern part of this ward are under the jurisdiction of Doublemooring Police Station of Chittagong City Corporation and the western part of Haliashahar Police Station. The coordinates of this ward is 22°19'35"N 91°48'7"E. According to the population census of 2011 it has a population of 1,08,513. Among them 56,606 are male and 51,907 are female. Total families are 22,912.

3.3 Demographic Information

The metro area population of Chittagong in 2021 was 5,133,000, a 2.25% increase from 2020. The metro area population of Chittagong in 2020 was 5,020,000, a 2.14% increase from 2019. Currently, 172,219 people are slum dwellers according to the Bangladesh Bureau of Statistics, 2021. In the event of rapid development, a high growth rate of population may be anticipated, which will cause rise of population to 8.82 million by the year 2032 Under medium growth rate, the population is estimated as 798 million, while the population under low growth rate is estimated at 6.02 million for the year 2032.

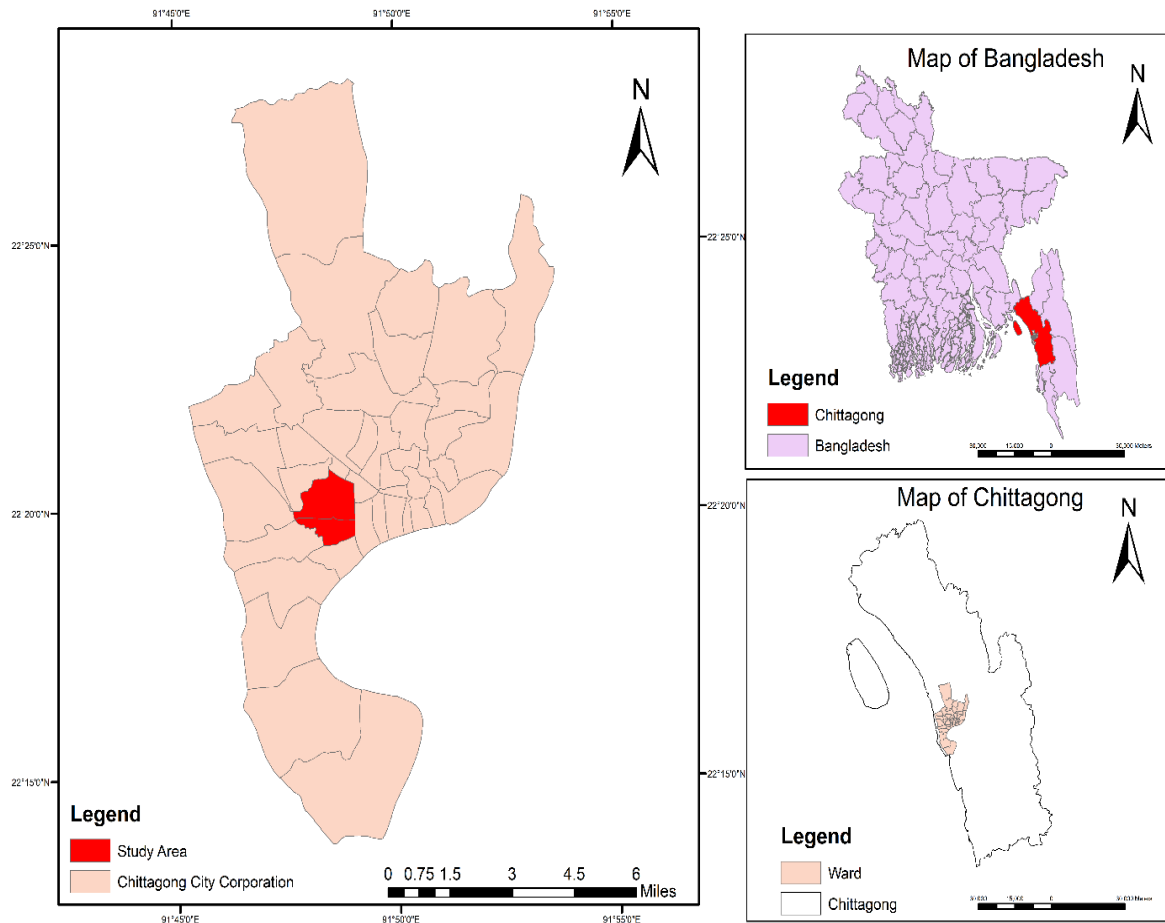


Fig.3.1: Study Area

3.4 Income Level

The Economy of Chittagong is reflected in its gross state product of \$116 billion in PPP terms ranking, second in size only behind Dhaka in Bangladesh. The nominal GDP of Chittagong metropolitan city is \$43 billion (The Chittagong Chamber of Commerce & Industry, 2022). Chittagong metropolitan City is the second largest metropolitan city in Bangladesh. Chittagong is the export hub of Bangladesh, accounting for more than \$20 billion in exports. Chittagong is the commercial hub of country. Chittagong is also home to the busiest port in the country.

3.5 Transportation

As Chittagong is a port city and the main commercial capital of Bangladesh. To ensure port productivity and competitiveness, efficient transport connectivity to/from the hinterland is important. The transportation system should be efficient and well developed in order for

the port to function as a gateway. To this end, the existing transport infrastructure is insufficient. Issues include traffic congestion, flooding, roads and bridges in poor condition, clogged canals and drains etc. (LGED, 2014).

3.6 Land Use

The major part of the city is occupied for residential purpose. The residential areas have ordinarily been assumed a neighborhood character. Each and every region has their distinguished characteristics, reminiscent of population density, house types, road network, utilities, and social classification, which have been developed either in a planned method or as spontaneous unplanned development. Chittagong has a long tradition of trade and commerce because of its locational advantage to sea port (M Shahidul Islam, 2012). Commercial land use can be characterized as mixed types which includes areas of retail and wholesale business, either formal or informal. Some sporadic commercial areas are also seen such as Lalkhan Bazar, Station Road, Dampara, Kattali, Bahaddarhat and Older parts of the city. These commercial areas are the heart of the city because here every day various types of commercial and administrative activities are dealt with. The traffic circulation is also higher in these areas than any other places of the city. The traditional wholesale business centers of the city are at Sadarghat, Khatungan and Chakktai, which have been developed without any systematic planning. To accommodate the major financial institutions and relevant offices with modern trade and commerce activities, CDA has developed Agrabad as a designated commercial area. This commercial area today has the Stock and Exchange Center, office of the Chamber of Commerce and Industries, and corporate branches of nearly all major banks. Although Agrabad is specifically meant for providing office spaces for commerce and trades, many buildings are currently being used for garments and other factories, which undermines the status of this commercial area (M Shahidul Islam, 2012). Chittagong city is increasing rapidly since independence, the city boundaries had been shifted and extra new lands are included inside city limit. Greater than 25% of city land is presently used as agricultural lands, commonly located in the Kalurghat field in the northeast, the foothills areas in the west and Patenga field within the south. However, with the gradual enlargement of non-farmed activities in these agricultural lands without following any progress guiding principle, city growth in these lands are taking

location is an awfully haphazard approach, principally for residential purposes (M Shahidul Islam, 2012).

3.7 Road Network

In Chittagong, wide roads exist throughout the city. There are various bus systems operated by public & private operators as well as taxis & CNG. Rickshaws are common as well as Dhaka & Chittagong highway is a major arterial highway & is the only way to access the city from Dhaka via land. The main roads are two -lane highway which are extremely congested in general. The CDA has been implementing widening of the major roads, as well as rehabilitation of damaged pavements including footpaths. The nature & extent of transportation services is influenced by the characteristics of road system which is inadequate in terms of both condition & capacity. Where roads are poor, most transport operators are deterred from providing service, while good road system is likely to encourage the development of bus services. But roads inside CCC are characterized by poor condition. The road network within the CCC can be classified into three distinctive categories These are Primary, Secondary Road network of CCC area.

3.8 Primary Road

Primary Roads are roads of continuous traffic flow including intra urban & inter urban Public Transport roads. Mainly this type of road is appropriate for bus operation(Abreha, n.d.). Width of the road varies from 70-80m. The primary roads in Agrabad are Sheikh Mujib Road, Port connecting Road & Dhaka Trunk Road.

3.9 Secondary Road

Roads that serve a lower level of intra urban mobility than the main arterial street & connect adjoining areas (Abreha, n.d.). The width is average 30-40m. Agrabad Access Road, 2 no Road, Halisahar Road, Faye Ahmed Road etc are access roads.

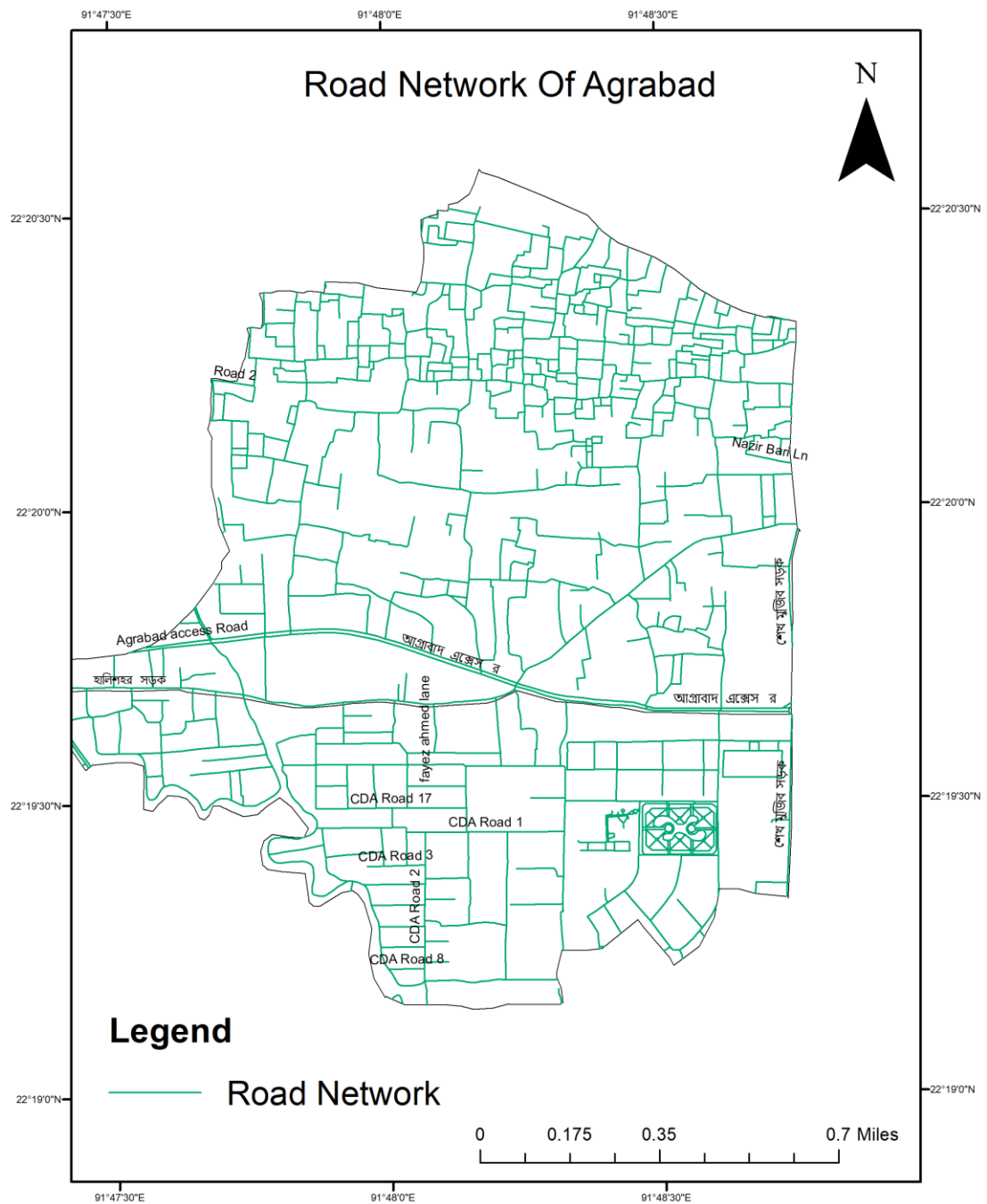


Fig.3.2: Road network of Agrabad

Chapter -4

Methodology

4.1 Data required

<u>Primary Data</u>	<u>Field Survey</u> • Field observation • Match the analytical evaluation with the actual field (After analysis)
<u>Secondary data</u>	• GIS datasets of infrastructures & roads. • BBS Data

4.2 Determination of sample size

The questionnaire survey will be conducted on specific sample size for this study. Sample size can be determined into two ways, probability sample and non-probability sample. In academic and government survey research, probability sampling is a standard procedure(Thompson, 2012). A thorough questionnaire has to be prepared to gather information relevant to the aims and objectives of the research which included questions evaluating every aspect of the mode choice of work trip making people in Agrabad, Chittagong. It will also include open-ended and close ended questions and way of data collection. To get proper information about study area it is needed to determine an appropriate sample size to bring the overall scenario of study area. Sampling means collecting data from a selected part of the total population(Wileyplus, 2001). This is an important portion of the research because if the sample number is not appropriate, it is difficult to find actual scenario of the study area. Total number of students and teachers in Chittagong city is about 5lac (BBS, 2013). The total number of buildings of Ward 27 (South Agrabad) is 2,908 (CDA, 2009). To conduct socio-economic & trip information survey, a sample of 550 workers have to be collected by simple random sampling procedure keeping the confidence level at 95%. Level of precision is taken 5% which means the error percentage is 4%.

The formula for sample size determination will be the following:

The sample size is to be determined by the formulas below (R A Johnson, 1977).

$$n_o = z^2 pq / e^2 \dots\dots\dots(1)$$

where,

n_o = Crude sample size

Z=Confidence level (95% confidence level, Z=1.96) (DeVore, 2017)

P=Target population

P+q=1, therefore q=0.5

e= Desired error which is 0.04

$$n_o = z^2 pq / e^2$$

$$n_o = (1.96)^2 * 0.5 * 0.5 / 0.04^2 = 600.25$$

$$n = n_o / 1 + n_o / N \dots\dots\dots(2)$$

where,

n=sample size

n_o = Crude sample size= 600.25

N=Population size= 1,75,268 (Wikipedia)

$$n = n_o / 1 + n_o / N = 600.25 / (1 + 0.0034) = 598.20$$

So, 600 sample has to be collected.

But due to the limited manpower & time period, 550 samples will be collected.

4.3 Identification of Significant Factors Affecting Mode Choice

After determining sample size, the survey will be conducted based on the questionnaire. In the questionnaire there will be some predetermined variables & these variables are found to influence the mode choice of work trips like gender, age, income, household number, origin, destination, length, time, fare of the mode, safety issue etc. The Multilinear Regression Analysis will be used. This method determines the independent variables influencing the dependent factors positively or negatively.

The following equation will be used:

$$Y = A + B_1 X_1 + B_2 X_2 + \dots + B_n X_n \dots \dots \dots (3)$$

Here,

Y = Dependent Variable;

X_i = Independent Variable (e.g. vehicle ownership, household ownership, walking time to & from stops, age)

A & B are parameters.

4.4 Determining The Effect of The Variable on Mode Choice

After-interpreting The Relationship of the Variables, The Utility Among the Transport Modes Will be Chosen by Using a Mathematical Model Named Multinomial Logit Model. It is the Simplest form of Mode Choice, where the travel choice between three or more than three Modes will be Made. The Model Can Be Expressed by the Following Equation:

$$c_{ij} = a_1 t_{ij}^v + a_2 t_{ij}^w + a_3 t_{ij}^t + a_4 t_{nij} + a_5 F_{ij} + a_6 \phi_j + \partial \dots \dots \dots (4)$$

Where, t_{ij}^v = in vehicle travel time between i and j

t_{ij}^w = walking to and from stops

t_{ij}^t = waiting time at stops

F_{ij} = fare charged to travel between I & j

Φ_j = parking cost

∂ = parameter

4.5 Determining Factors Affecting Travel Satisfaction

In this step factors which are affecting the travel satisfaction of the workers are determined. This step actually determines the utility value of different satisfaction factor. For this research five factors which are cost comfort, accessibility, safety, security is taken to determine the utility and the importance value of the factors.

$$U = U_{\text{cost}} + U_{\text{comfort}} + U_{\text{Acc}} + U_{\text{safety}} + U_{\text{security}} + C \dots \dots \dots (5)$$

Where, U=Overall Utility

U_{cost} , U_{comfort} , U_{Acc} , U_{safety} , U_{security} is the utility of cost, comfort, accessibility, safety, security respectively.

C= constant

With the factors utility value of each respondent, his or her factor weight can be gotten with the following formula

$$V_{ij} = \frac{\max U_{ij} - \min U_{ij}}{\sum (\max U_{ij} - \min U_{ij})} \dots \dots \dots (6)$$

Where, V_{ij} = j's factor weight of respondent i.

U_{ij} = j's factor utility value of respondent i.

Then each factor's weight can be got through composting all the respondent's weight, that is:

$$V_i = V_{ij} / \sum_j V_{ij} \dots \dots \dots (7) \text{ (Liu et al., 2011)}$$

4.6 Identifying the Trip Pattern of the Worker

Identification of trip pattern is important to know which road is mainly used by the people making trips. Different mode uses different road. It is identified by network analysis in ArcGIS. There is a tool for shortest route in network analysis. By this tool the trip route which is being used by the worker can be identified. A hypothesis is taken that worker usually take the shortest route to reach their working place.

4.7 Conjoint Analysis

Stated preferences method & choice based conjoint analysis can be used to analyze customer preferences to assess quality & public transport service contracts (Gatta & Marcucci, 2007). Estimating the passenger's evaluation of different service features & calculating a service quality index. This is followed to measure service quality in local public transport. Moreover, according to many researchers stated preference techniques have greater advantages over revealed preference techniques based on actual choices. Because the individual can make more than one transportation choice & can be presented with trade-offs rather than dominated choices (Zotti & Danielis, 2004).

4.8 Satisfaction Scoring

A capitalization weighted index is a form of index with distinct constituents that are prejudiced according to their total observer value. The higher components carry higher percentage of weightings while the smaller components in the index have smaller weights. This type of index is known as value weighted index. Multiplying each components value by its total owing shares to attain at the total factor value. Then, percentage of this value to the total factor value of all the index constituents gives the weightage of the factor in the index (Hayes Adam, 2022).

4.9 Findings & Recommendations

It is the finishing step of the working procedures & term of taking decisions. After completion of data analysis, the major findings will be summarized. Relevant information based on the findings will also be provided.

4.10 Outcome of the Research

- From this analysis find out the developed model can be used in travel demand analysis to develop efficient transport policies.
- To find out the choice behavior of employed people.
- To find out the most suitable mode choice of employed people used for work trips in Agrabad.

4.11 Methodology Flowchart

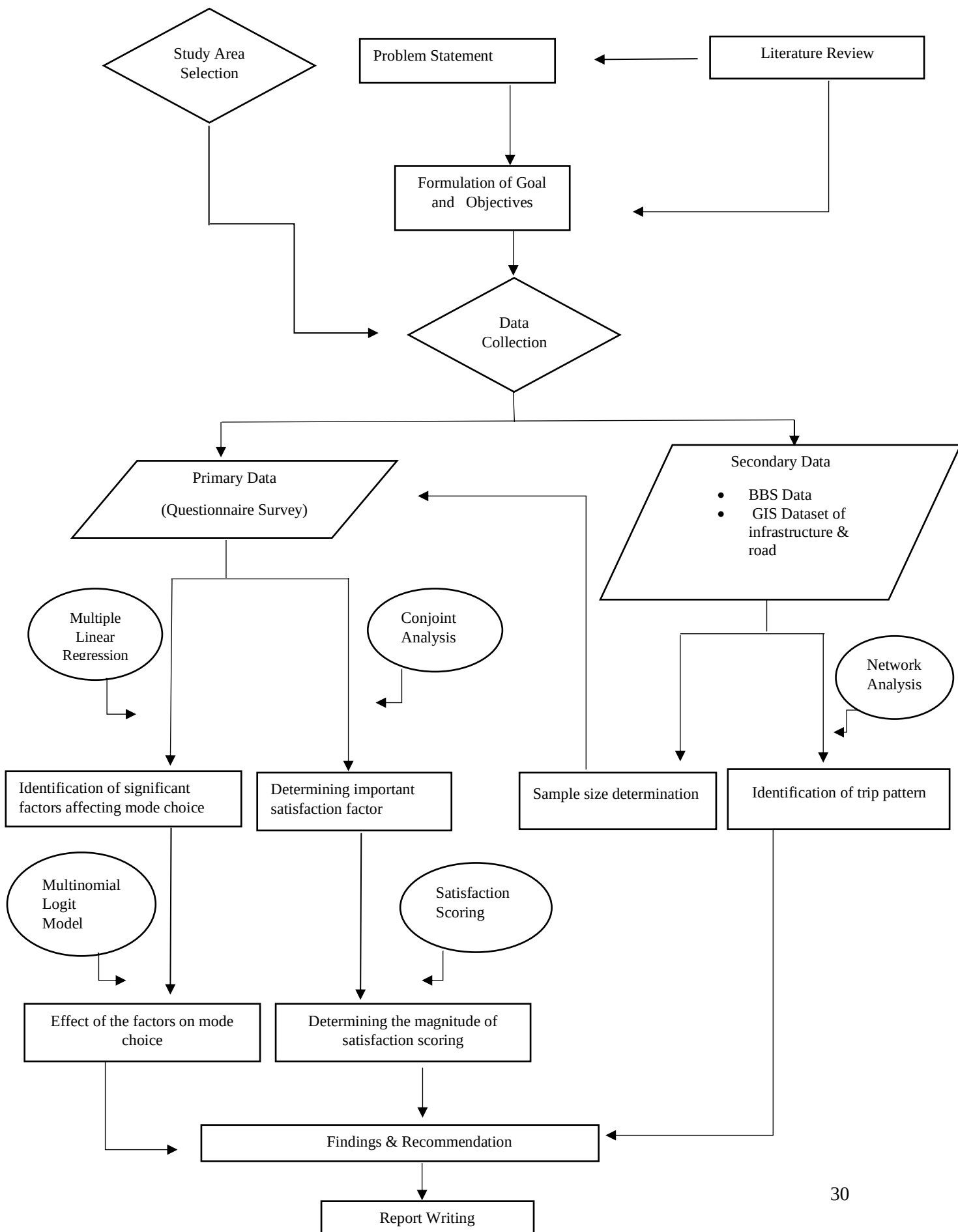


Fig.4.11 Methodology Flowchart

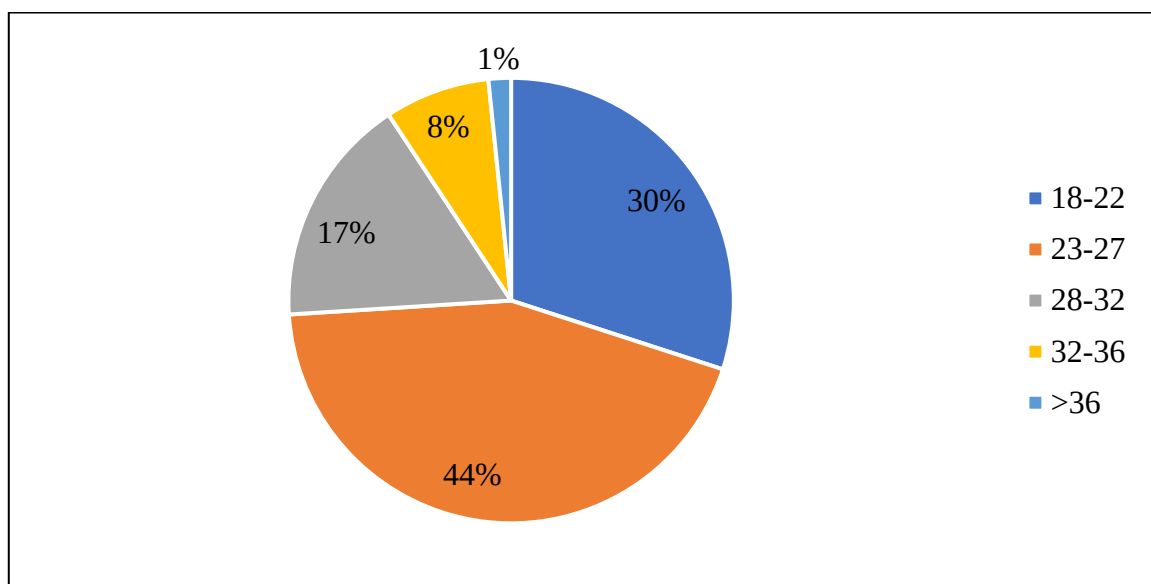
Chapter -5

Data Analysis

This chapter in this study discuss about the data analysis and findings and deals with the current attitude and travel behavior of the working trips along with the travel satisfaction level of them Also find out the factors that influence the mode choice behavior and develop Regression Model and Multinomial Logit Model by using the data collected through the questionnaire survey and finally satisfaction level using conjoint analysis.

5.1 Socio Economic Aspect

5.1.1 Age



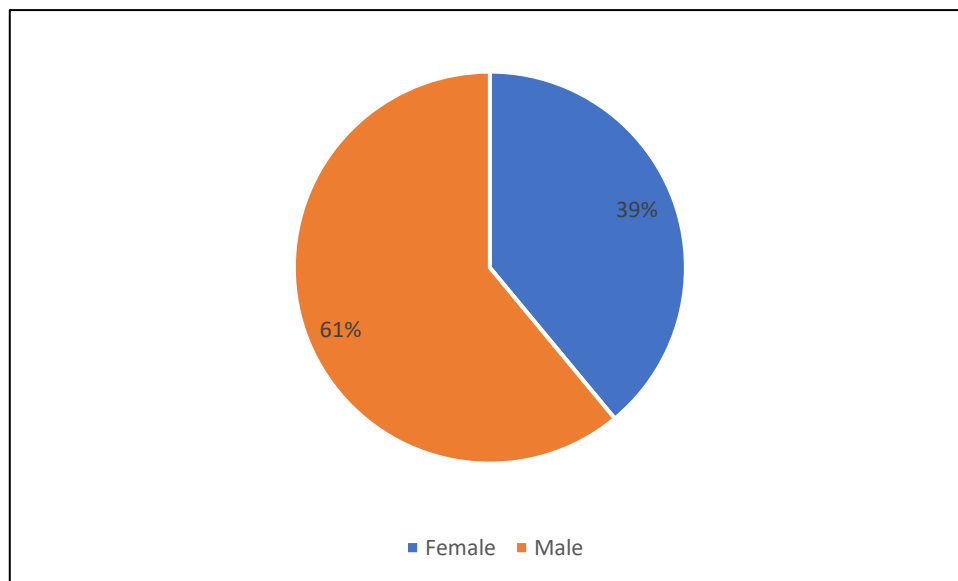
Source: Field Survey, 2022

Figure no. 5.1: Age of the respondent

Figure 5.1 is showing the percentage of respondent of different age group. It can be seen that most of the people are between 23 to 27 years old which is 44% of the total employed people. The second most age group is between 18 to 22. This indicate that a good number of young people is engaged in work trip. These people have actually low or so-so level of educational qualification (which is analyzed the following part) but have to manage their family. This made them engaging into this sector mainly. 17% of respondents are 28-32 years old, 8% of people are between 32 to 36 years old and only 1% of people is more

than 36 years old. This analysis is showing that with getting old, the number of workers is actually decreasing. Reason behind this can be the hard work of this sector.

5.1.2 Gender

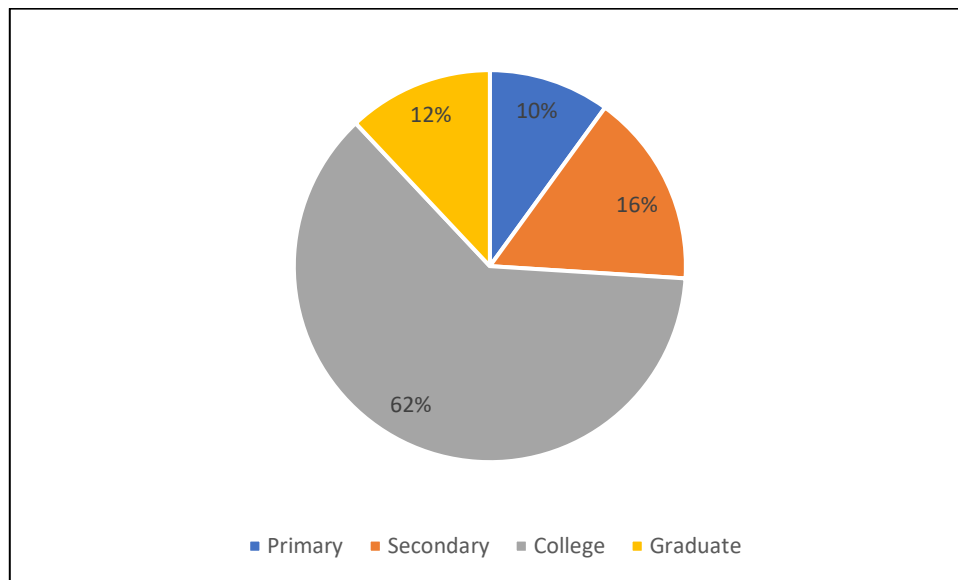


Source: Field Survey, 2022

Figure no. 5.2: Gender of the respondent

The figure 5.2 is showing the percentage of gender of employed people working in different sectors in Agrabad. From the figure it can be seen that most of the respondents are female.

5.1.3 Education



Source: Field Survey, 2022

Figure no. 5.3: Education of the respondent

From the figure 5.3 it can be seen that most of the workers are only college pass which is about 62% of the total worker. Only 10% of them passed primary education, and 12% of the respondents are graduate, working as officer in factories. Education is an important factor for the workers for getting promotion into higher designation. Without having good educational qualification, it is very rare for the workers getting promotion else they have a good working experience. As mainly working-class people are been surveyed it is found that most of them have only passed college level.

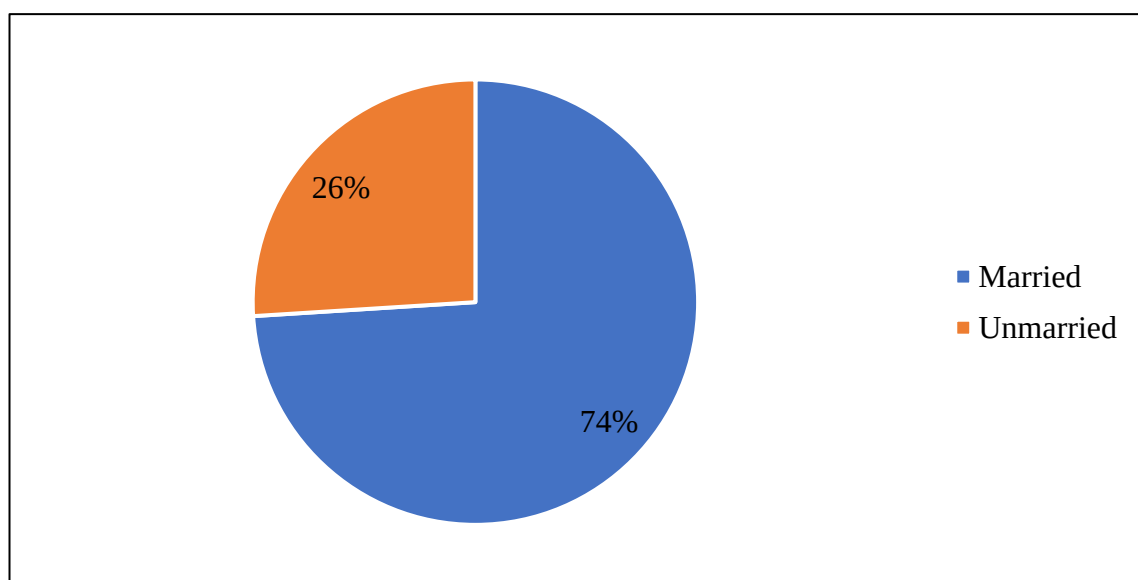
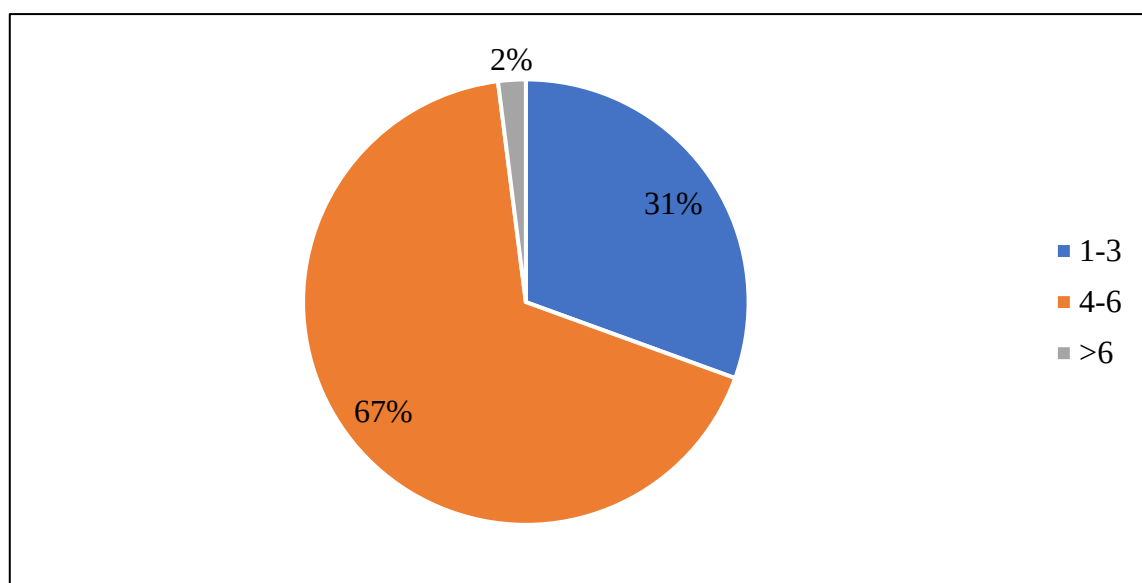


Figure no.5.4: Marital Status of the respondent

Marital status of the employed people is shown in the figure above. This is showing that 74% of the workers are married and 26% of them are unmarried. It is seen in figure 5.4 that most of the worker are more than 22 years old it means that they can legally conduct marriage. The rest 26% are such worker who either not properly aged for marriage or have not conducted marriage due to other problems.

5.1.5 Number of Household Member

Household number is very important & it may affect the mode choice as with the increase of household member the expenditure also increases. This can be responsible to choose the mode for both working and non-working purpose.



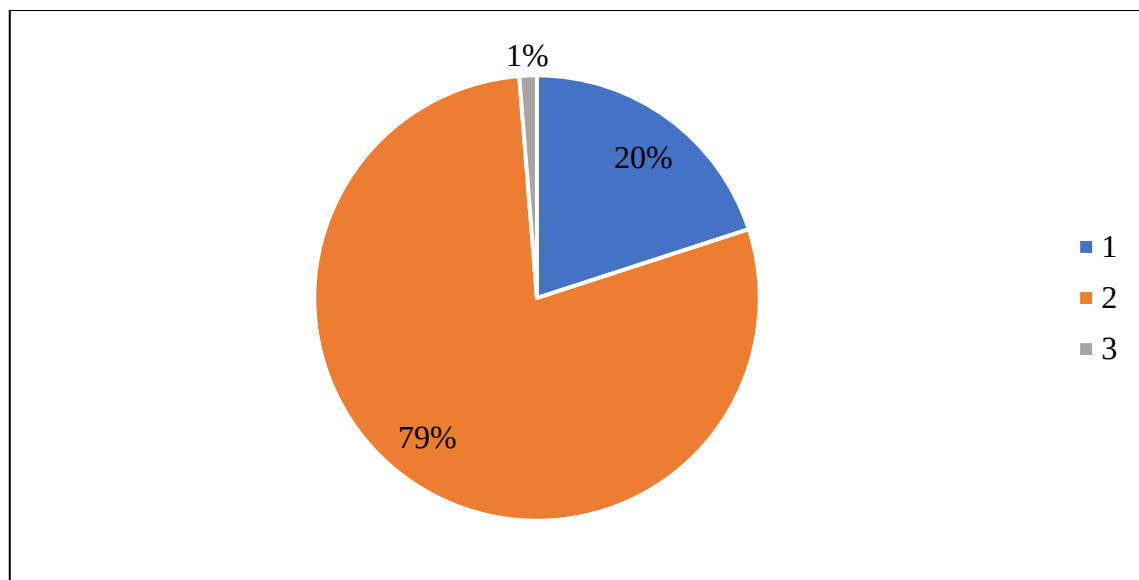
Source: Field Survey, 2022

Figure no. 5.5: Number of Household Member

It is found from figure 5.5 that most of the family have 4-6 household member which is 67%. With the increase of household member another thing also increases which is the number of earning member. As to maintain bigger family more money is needed that is why more people get engaged in the working sector. 31% of household is found to have 1 to 3 members. These are nuclear family who actually live with their husband/wife and one child. The existence of family having more than 6 household members are also found rarely but the percentage is very small than other which is about 2%.

5.1.6 Number of Earning Member

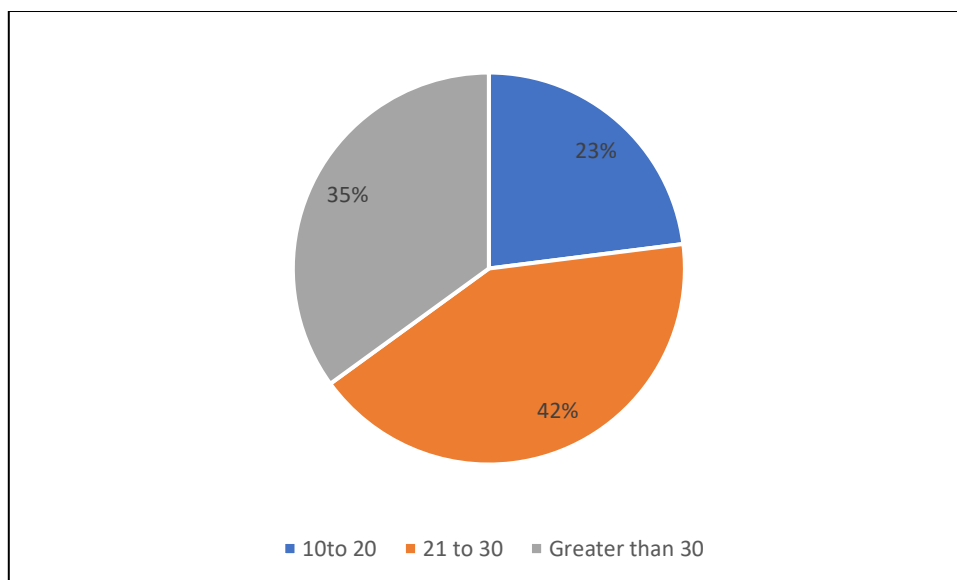
Figure no. 5.6 is showing the percentage of number of earning member of each household engaged in multiple industries. It can be seen that 79% of household actually have 2 earning member. It is known from the personal interview with officials that the normal salary is between 21-30 thousand taka per capita per month. This amount of money fall very short to run a family in even so-so way. That is why in maximum family 2 people are getting engaged in this corporate sector to maintain their family in a proper way. The second most percentage is found for number of earning member on each household in 1 which is 20% of total. This represents mainly little family sized household and mainly of those group whose income is better than other. Only 1%of household is found in which there are 3 earning member. Their income is mainly between 21 to 30 thousand and sometime more than 30 thousand which is analyzed in the cross tab between number of earning member and household income.



Source: Field Survey, 2022

Figure no. 5.6: Number of Earning Member

5.1.7 Income (in thousand)



Source: Field Survey, 2022

Figure no. 5.7: Income Status

The chart showing above is describing the income status of the people employed in work trip. It is seen that 35% of people's household income is greater than 30 thousand. It is very normal because in the research from the collected data it is seen that most of the family's working member is equal or more than 2 persons. The lowest salary of employed people is 10 to 20 thousand taka which is known from a worker. And 42% people's income is less than or equal to 30 thousand. These are actually such type of people who either is the only one earning member of the household.

5.1.8 Education vs Income (in thousand) Crosstab

Table no. 5.1: Cross tabulation between Education and Income

			Income			Total
			10-20 Thousand	21-30 Thousand	>30 Thousand	
Education	Primary	Count	0	8	1	9
		% within Education	88.9%	0.0%	11.1%	100.0%
		% within Income	0.0%	2.0%	2.6%	1.6%
		% of Total	0.0%	1.5%	0.2%	1.6%
	Secondary	Count	10	54	24	88
		% within Education	11.4%	61.4%	27.3%	100.0%
		% within Income	8.8%	13.6%	2.6%	16.0%
		% of Total	1.8%	9.8%	4.4%	16.0%
	College	Count	104	334	13	451

		% within Education	23.10%	74.10%	2.90%	100.00%
		% within Income	91.20%	84.30%	33.30%	82.10%
		% of Total	18.90%	60.80%	2.40%	82.10%
	Graduate	Count	0	0	1	1
		% within Education	0.0%	0.0%	100.0%	100.0%
		% within Income	0.0%	0.0%	61.5%	0.2%
		% of Total	0.0%	0.0%	0.2%	0.2%
	Total	Count	114	397	39	550
		% within Education	20.8%	72.1%	7.1%	100.0%
		% within Income	100.0%	100.0%	100.0%	100.0%
	% of Total	Count	20.8%	72.1%	7.1%	100.0%

The table no 5.1 is showing a crosstab between education and income of the worker. It is obvious that with better education comes better income. This is depicted in the cross tab. It can be seen that 23.1% of workers who are college pass are making an income of less than 20 thousand which is actually 91.1% of the total worker making such amount of income. The 20-30 thousand income earner are also mainly college pass worker who are 84.3% of the total such amount earner. 61.15% of greater than 30thousand taka income earner is from secondary pass worker. The income of college pass workers is here confined between 20-30thousand. This is because college pass people's tendency to work as a worker is very low.

5.1.9 Gender vs Income Crosstab

This table 5.2 is showing the cross tabulation between gender and income of the workers. It is found that the number of female workers is higher than male workers. It is found from the table that among male workers most of them are getting a household income of 10-20thousand which is about 66.2%. A very little percentage of worker is getting 20 thousand taka per month which is only 13%. On the other hand, the scenario is quite same for female

workers. 72% of them is earning 11-20 thousand takas monthly and only 6.7% of female are earning more than 20 thousand taka.

Table no. 5.2: Cross tabulation between Gender and Income

			Income class			Total
			10-20 Thousand	20-30 Thousand	>30 Thousand	
Gender	Male	Count	70	255	9	334
		% within Gender	21.0%	76.3%	2.7%	100.0%
		% within Income class	60.9%	64.1%	75.70%	60.7%
		% of Total	12.7%	46.4%	1.6%	60.7%
	Female	Count	45	143	28	216
		% within Gender	20.80%	66.20%	13.00%	100.00%
		% within Income class	39.10%	35.90%	24.3%	39.30%
		% of Total	8.20%	26.00%	5.10%	39.30%
Total		Count	115	398	37	550
		% within Gender	20.9%	72.4%	6.7%	100.0%
		% within Income class	100.0%	100.0%	100.0%	100.0%
		% of Total	20.9%	72.4%	6.7%	100.0%

Now, within income class it is observed that 39.10% of female workers are earning between 10 -20 thousand taka and 64.1% of male are getting 20-30 thousand taka. In these income class male percentage is much higher than female. In the income class of more than 30 thousand the male percentage is far better than female. 75.7% of male are earning more than 30 thousand taka whereas 24.3 % of female are making such amount of household income. From the table it is clear that, female workers have majority in low- and medium-income class whereas male are getting more income than female.

5.1.10 Number of Earning Member vs Income Class Crosstab

Table 5.3 is demonstrating the cross tabulation between number of earning member and income class of the employed people. From the table it is found that with the increase of earnings member the amount of household income also increases but not in all time.

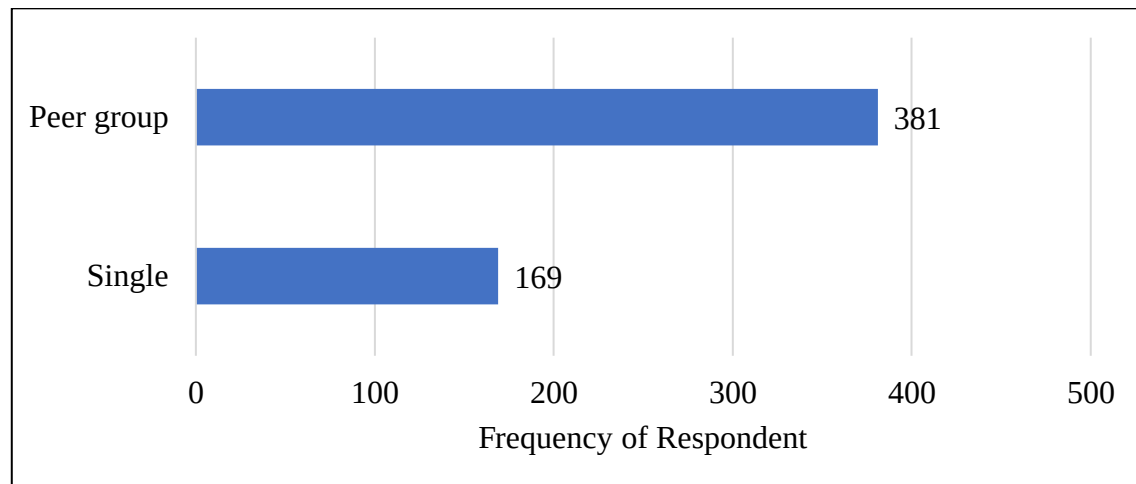
Table no. 5.3: Cross tabulation between Earning member and Income class

			Income class			Total
			10-20 Thousand	21-30 Thousand	>30 Thousand	
Earning member	1	Count	84	26	0	110
		% within Earning member	76.4%	23.6%	0.0%	100.0%
		% within Income class	73.7%	6.5%	0.0%	20.0%
		% of Total	15.3%	4.7%	0.0%	20.0%
	2	Count	30	365	38	433
		% within Earning member	6.9%	84.3%	8.8%	100.0%
		% within Income class	26.3%	91.9%	97.4%	78.7%
		% of Total	5.5%	66.4%	6.9%	78.7%
	3	Count	0	6	1	7
		% within Earning member	0.0%	85.7%	14.3%	100.0%
		% within Income class	0.0%	1.5%	2.6%	1.3%
		% of Total	0.0%	1.1%	0.2%	1.3%
Total		Count	114	397	39	550
		% within Earning member	20.7%	72.2%	7.1%	100.0%
		% within Income class	100.0%	100.0%	100.0%	100.0%
		% of Total	20.7%	72.2%	7.1%	100.0%

It is already found that family having two earning members is much higher than 1 or 3. From the table it can be seen that, family having 1 earning member is actually getting 10-20 thousand household income and the percentage of such worker is 73.7%. 26.3% of household is earning 10-20 thousand taka per month in which there are 2 earning members. No household are found to earn between 10-20 thousand taka if their earning member is 3.

84.3% of household having 2 earning member is getting household income between 20 to 30 thousand taka which is the most among number of earning members.

5.1.11 Travel Companion



Source: Field Survey, 2022

Figure no. 5.8: Travel Companion

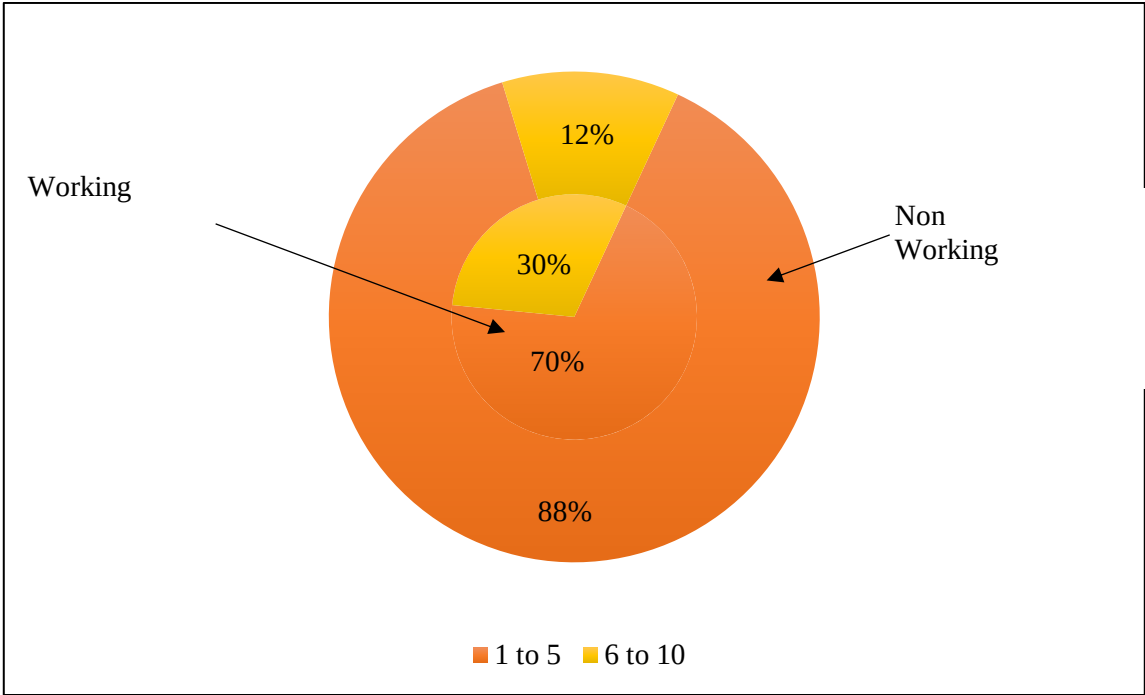
Travel companion means that the companion pattern of the worker during conducting work trip. It is mainly divided into two types single and peer group. From the analysis it is found that most of the workers actually go to their working place in peer group. It happens so as from the survey it is found that the employed people mainly live in clustered way. So, during the time to going to their working place they go together. It actually makes them feel better as walking with someone if far better than walking alone. Besides they feel safe also.

5.2 Travel Behavior Aspect

5.2.1 Length of the Trip

Figure no. 5.9 is demonstrating the length of distance travelling for working and non-working purpose. It has been found that the trip time is usually not very much high and the mode mainly used by the worker for working and non-working purpose is walk it is obvious that the length of their trip shall not be very high which is actually found in figure. It can be seen that for working purpose the workers don't actually go very further from their household. 70% of worker's working place trip length is in between 1 to 5 KM. Rest actually use motorized transport for their working trip. On the other hand, the scenario of trip length for non-working purpose is much more on lower trip length's side. Only 12%

of worker travel 6 to 10 KM distance from their house for non-working purpose. It is been said by the workers that they are not actually tend to travel or do recreational activity on their leisure time as their income is limited.



Source: Field Survey, 2022

Figure no. 5.9: Length of the Trip (in KM)

5.2.2 Mode of Transport

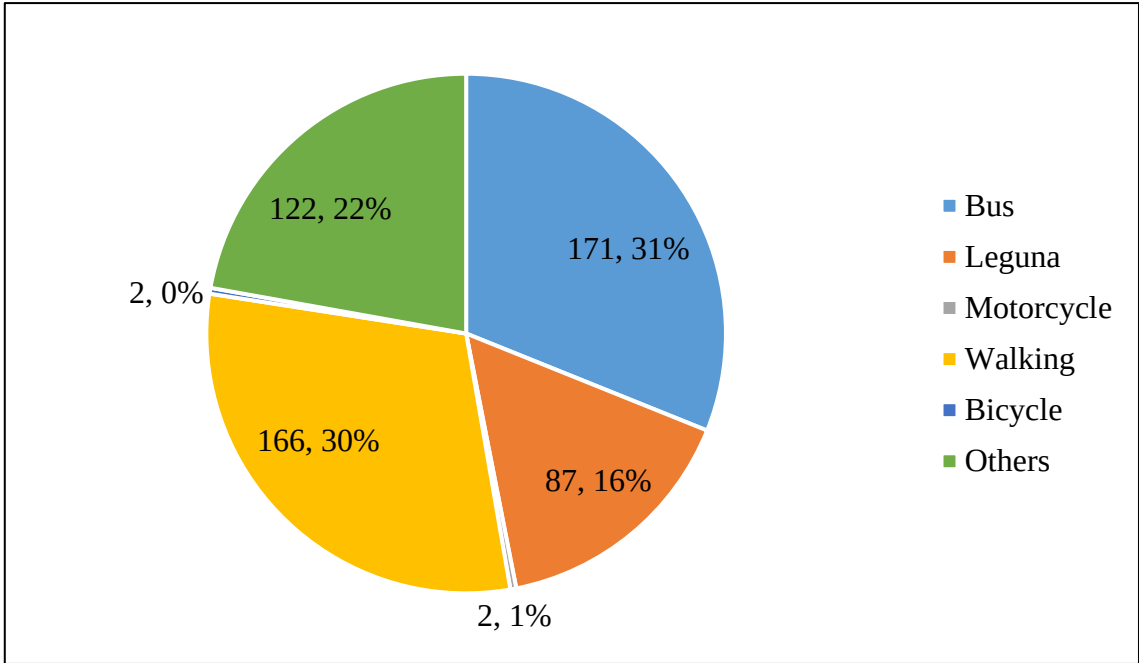
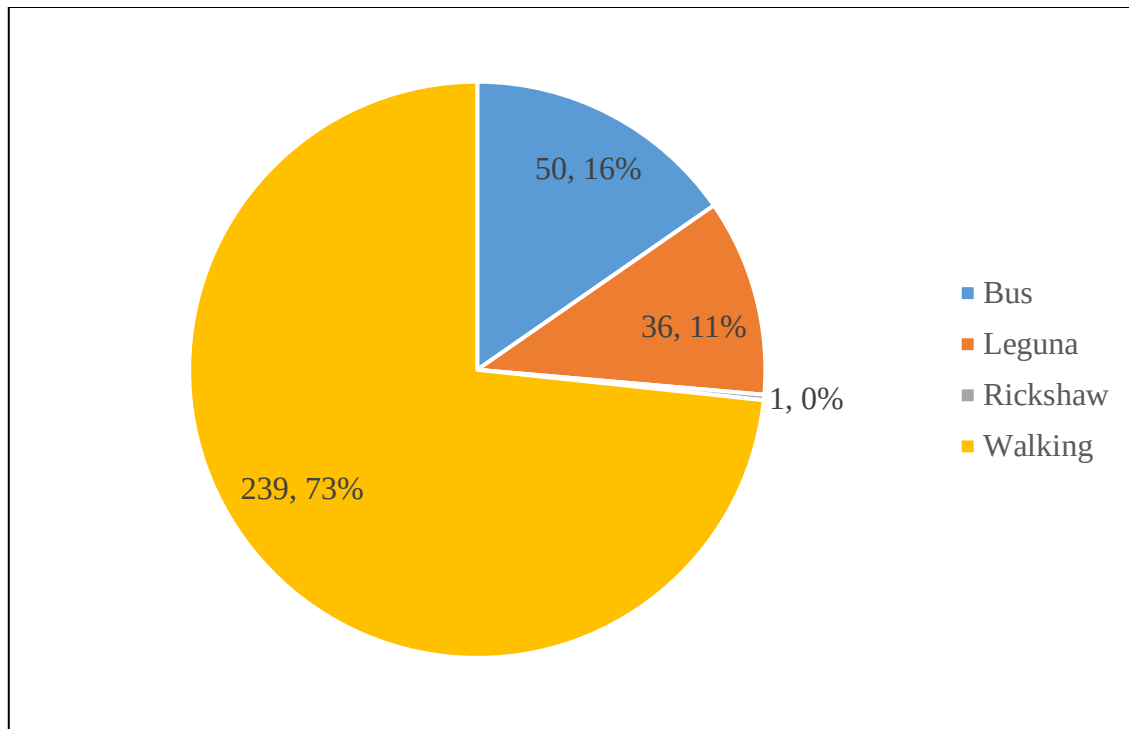


Figure no. 5.10: Mode of Working Trip

Source: field survey,2022



Source: Field Survey, 2022

Figure no. 5.11: Mode of Non-Working Trip

Figure 5.10 and 5.11 is showing the number and the percentage of different mode used for work and non-work trip respectively. It can be seen that for working trip most of the people use bus as their mode of travelling. Second most is walking mode. Reason behind choosing these modes is less fare which is analyzed with multinomial logit model and shown in the models of mode choice for work trip. Besides, other which is actually the institutional vehicle is the third most mode used for the work trip. This is mainly used by the persons whose income is of mid to high range and are more concerned about the safety, security and comfort issue and so as leguna. There is some people who has their own vehicle which is motorcycle and bicycle. 2 of them have bicycle and 2 have motorcycle. On the other hand, for non-working trip they mostly use walk as their main mode. People mostly go to shop, market, bazar which is located in their walking distance. Some of the people go to park, distanced market, zoo, beach, cinema for recreational activity mostly using bus and then leguna. Very little amount of people use rickshaw as the fare is very high in it.

5.2.3 Mode Choice vs Income Class Crosstab

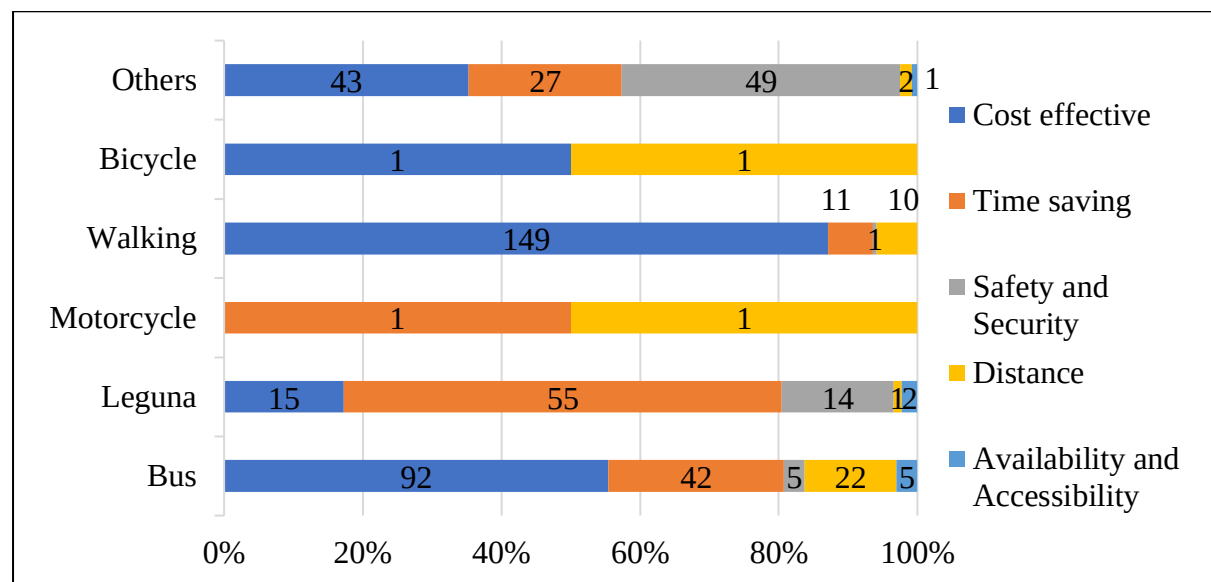
Table no. 5.4: Cross tabulation between Working Trip Mode and Income

			Income Class			Total
			10-20 Thousand	21-30 Thousand	>30 Thousand	
Mode	Bus	Count	38	112	16	166
		% within Mode	22.9%	67.5%	9.6%	100.0%
		% within Income Class	33.3%	28.2%	41.0%	30.2%
		% of Total	6.9%	20.4%	2.9%	30.2%
	Leguna	Count	23	55	9	87
		% within Mode	26.4%	63.2%	10.3%	100.0%
		% within Income Class	20.2%	13.9%	23.1%	15.8%
		% of Total	4.2%	10.0%	1.6%	15.8%
	Motorcycle	Count	0	2	0	2
		% within Mode	0.0%	100.0%	0.0%	100.0%
		% within Income Class	0.0%	0.5%	0.0%	0.4%
		% of Total	0.0%	0.4%	0.0%	0.4%
	Walking	Count	40	122	9	171
		% within Mode	23.4%	71.3%	5.3%	100.0%
		% within Income Class	35.1%	30.7%	23.1%	31.1%
		% of Total	7.3%	22.2%	1.6%	31.1%
	Bicycle	Count	1	1	0	2
		% within Mode	50.0%	50.0%	0.0%	100.0%
		% within Income Class	0.9%	0.3%	0.0%	0.4%
		% of Total	0.2%	0.2%	0.0%	0.4%
	Others	Count	12	105	5	122
		% within Mode	9.8%	86.1%	4.1%	100.0%
		% within Income Class	10.5%	26.4%	12.8%	22.2%
		% of Total	2.2%	19.1%	0.9%	22.2%
Total		Count	114	397	39	550
		% within Mode	20.7%	72.2%	7.1%	100.0%
		% within Income Class	100.0%	100.0%	100.0%	100.0%
		% of Total	20.7%	72.2%	7.1%	100.0%

This table no. 5.4 is showing the crosstab relationship with household income and mode choice of the worker. From the table it can be seen that 33.3% of people belong from income class of 10-20 thousand taka & they use bus as primary mode of transport. The high

income people also use bus as their main mode of transport. 23.1 % of people use leguna for work trip which is the second most used mode & walking is the third used mode here.

5.2.4 Relation between Working Mode of Transport and The Reason of Choosing Mode



Source: Field Survey, 2022

Figure no. 5.12: Relation between Working Mode of Transport and The Reason of Choosing Mode

There are several reasons for choosing different type of mode by the worker. Such five reasons were found through the questionnaire survey. The figure 5.12 is showing the main reasons of choosing different modes. At first for bus, it is found that 92 of the workers have said that they are using bus rather than other mode because it is cost effective. Usually workers who use bus, travel a longer distance. But the fare of bus is not so high which will be showing the following analysis. That's why for travelling a longer distance worker usually choose bus. In case of leguna the main reason is found time saving. The travelling time of leguna is lower than other modes as it does not have very high capacity and the tendency of stopping and waiting for the passenger here and there is lower than bus. So,

worker who prefer fast travelling prefer leguna. Motorcycle is used by only two of the workers surveyed and one of them has said that he is using motorcycle because of greater distance and another one is using because of less time. In walking 85% of the worker's walk mode using reason is the cost of trip. It is obvious that in walk mode no cost is required. That's why for shorter distance they usually walk. Like the motorcycle bicycle is also used by two of the workers but the reason is not same. One of the users has mentioned cost saving as the reason and another one has mentioned distance. In the graph other means the official vehicle. This is mainly used by the passengers because of its better safety and security.

5.2.5 Length of Working Trip vs Modes for Work Trip

Crosstabulation

Table no. 5.5: Cross tabulation between Length of Working Trip and Modes for Work Trip

			Length of Trip (in KM)		Total
			1-5	6-10	
Mode	Bus	Count	106	60	166
		% within Mode	63.9%	36.1%	100.0%
		% within Length of Trip	14.4%	35.7%	30.2%
		% of Total	19.3%	10.9%	30.2%
	Leguna	Count	55	32	87
		% within Mode	63.2%	36.8%	100.0%
		% within Length of Trip	27.7%	19.0%	15.8%
		% of Total	10.0%	5.8%	15.8%
	Motorcycle	Count	1	1	2
		% within Mode	50.0%	50.0%	100.0%
		% within Length of Trip	0.3%	0.6%	0.4%
		% of Total	0.2%	0.2%	0.4%
	Walking	Count	169	2	171
		% within Mode	98.8%	1.2%	100.0%
		% within Length of Trip	44.2%	1.2%	31.1%
		% of Total	30.7%	0.4%	31.1%
	Bicycle	Count	1	1	2
		% within Mode	50.0%	50.0%	100.0%
		% within Length of Trip	0.3%	0.6%	0.4%
		% of Total	0.2%	0.2%	0.4%

	Official Vehicle	Count	50	72	122
		% within Mode	41.0%	59.0%	100.0%
		% within Length of Trip	13.1%	42.9%	22.2%
		% of Total	9.1%	13.1%	22.2%
Total		Count	382	168	550
		% within Mode	69.5%	30.5%	100.0%
		% within Length of Trip	100.0%	100.0%	100.0%
		% of Total	69.5%	30.5%	100.0%

This table is showing for which distance worker use what mode. It means if worker have to travel which mode they are using and for shorter trip distance which mode is being used by them. It is found from the table that for shorter distance which is 1-5 KM worker mostly walk. The cross tabulation is showing that 44.2% of the workers usually walk when the length of the trip is between 1 to 5 KM. The second most used mode for such length is leguna. On the other hand, for travelling longer distance workers are mainly found to be using official vehicle. Reason behind using this has been found in figure no 5.5 which is safety and security. It is normal that for longer distance the chance of accident is high. As there is no safety issue in official vehicle that's why this is mostly used which is about 42.9%. Second mostly used mode for longer trip length is found bus.

5.2.6 Length of Non-Working Trip vs Modes Used Cross tabulation

Unlike the previous cross tab, the following one is about the non-working perspective. This cross tab is showing which mode is used by the worker in longer or shorter length for their non-working trip. For shorter trip length, like the working trip, walk is used but the percentage is very much higher than the previous one. About 81.3% of the worker walk to their destination. It is also found that worker usually walk shorter distance not the longer one as 97.9% of worker belong to the group among the total worker who generally use walk mode for their non-working trip. Rickshaw is found to be used by only one worker for the trip length of 1-5 KM. Worker usually do not use rickshaw as it is very much expensive. On the other hand, for longer distance worker use bus. But worker usually do not travel longer distance for non-working purpose.

Table no.5.6 Cross tabulation between Length of Non-Working Trip and Modes

			Length of Trip (in KM)		Total
			1-5	6-10	
Mode	Bus	Count	28	22	50
		% within Mode	56.0%	44.0%	100.0%
		% within Length of Trip	9.7%	57.9%	15.3%
		% of Total	8.6%	6.7%	15.3%
	Leguna	Count	234	5	239
		% within Mode	97.90%	2.10%	100.00%
		% within Length of Trip	81.30%	13.20%	73.30%
		% of Total	71.80%	1.50%	73.30%
	Rickshaw	Count	1	0	1
		% within Mode	100.0%	0.0%	100.0%
		% within Length of Trip	0.3%	0.0%	0.3%
		% of Total	0.3%	0.0%	0.3%
	Walking	Count	25	11	36
		% within Mode	69.4%	30.6%	100.0%
		% within Length of Trip	28.9%	8.7%	11.0%
		% of Total	7.7%	3.4%	11.0%
Total		Count	288	38	326
		% within Mode	88.3%	11.7%	100.0%
		% within Length of Trip	100.0%	100.0%	100.0%
		% of Total	88.3%	11.7%	100.0%

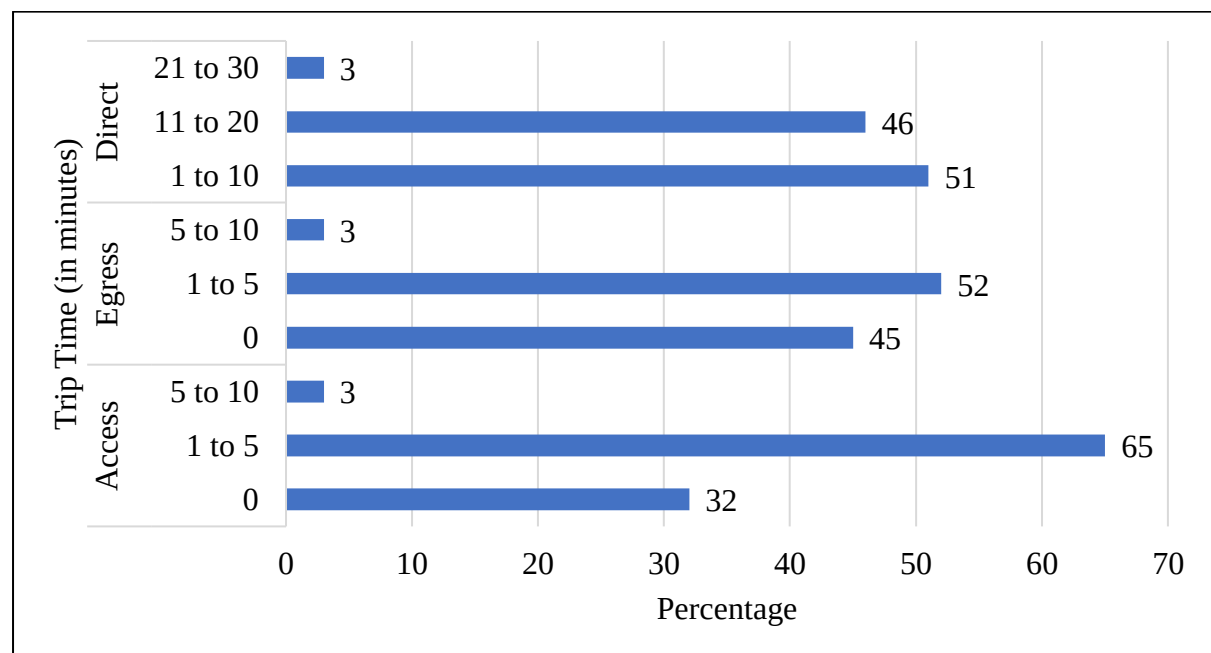
57.9% of respondents use bus for travelling to longer distance which is 6-10 km. For distance of 1to 5 km, they mainly use leguna which is for 1 to 5 km is 81.30% & walking is the third & last used mode.

5.2.7 Trip Time

Figure no. 5.13 and 5.14 is demonstrating the access, egress and direct trip time of working and non-working mode respectively. It is found from figure 5.13 that in some case there is no access and egress time. It happens because a good number of workers walk to their working place and in walking mode there is no access or egress time. The whole time is counted as the direct time. But for other mode there will be access and egress time as from house the worker needs to walk or use some other mode to reach to the place from where

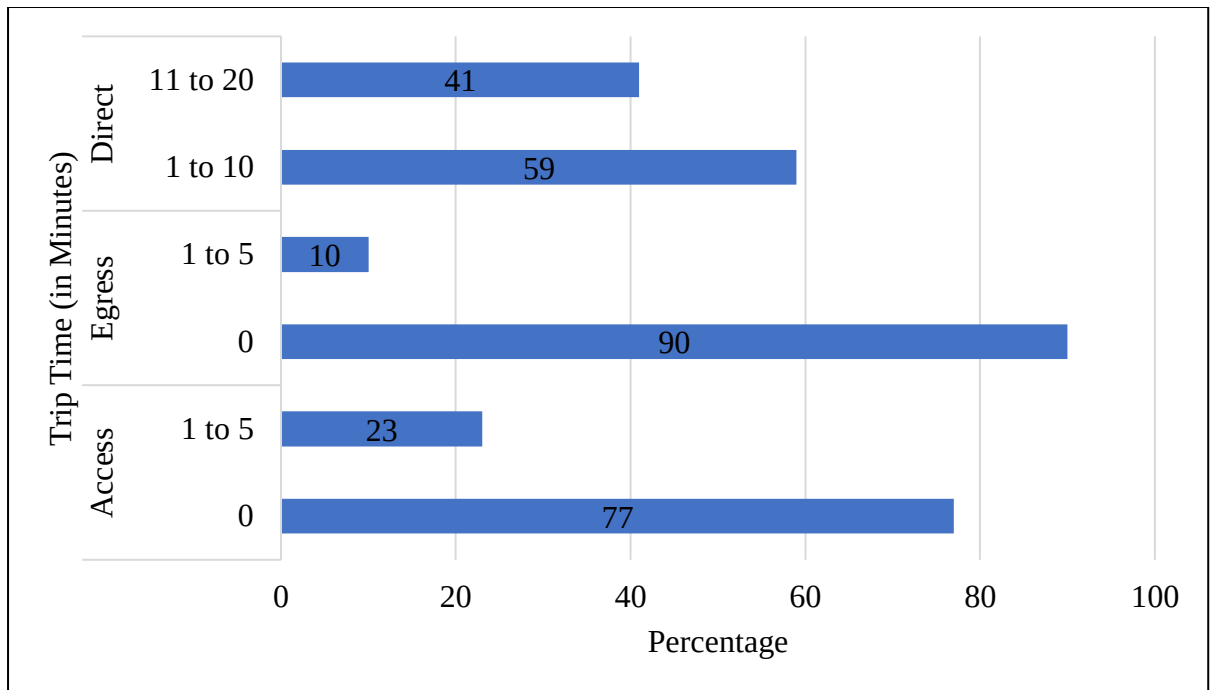
they will ride to the vehicle. For most of the worker this access time is between 1 to 5 minutes which is about 65%. The scenario is same for egress time too. 52% of the egress time is in between 1 to 5 minutes. It refers that, workers who do not walk to their working place are living in a house which is not so far from the main road. And also, the factory where they work are also not so very far from the main road. For direct time, it can be seen that most of the worker's direct time is in between 1 to 5 minutes. As they most of the worker trip length is between 1 to 5 KM and a good number of workers use motorized vehicle, it is obvious that the trip time shall not be very high. But walking takes a good amount of time and that's why 46% of worker's working trip time is in between 11 to 20 minutes.

For non-working trip time it can be seen in figure 5.14 that nonexistence of access and egress time is higher. It is shown in figure no 5.11 that most of the worker usually take walk mode for non-working mode and that's why for 77% and 90% of workers access and egress time do not exist respectively which is shown as 0 minutes in the figure. In the direct time as worker's do to travel longer distance for non-working purpose that's why the time is mostly between 1 to 10 minutes for 59% of workers.



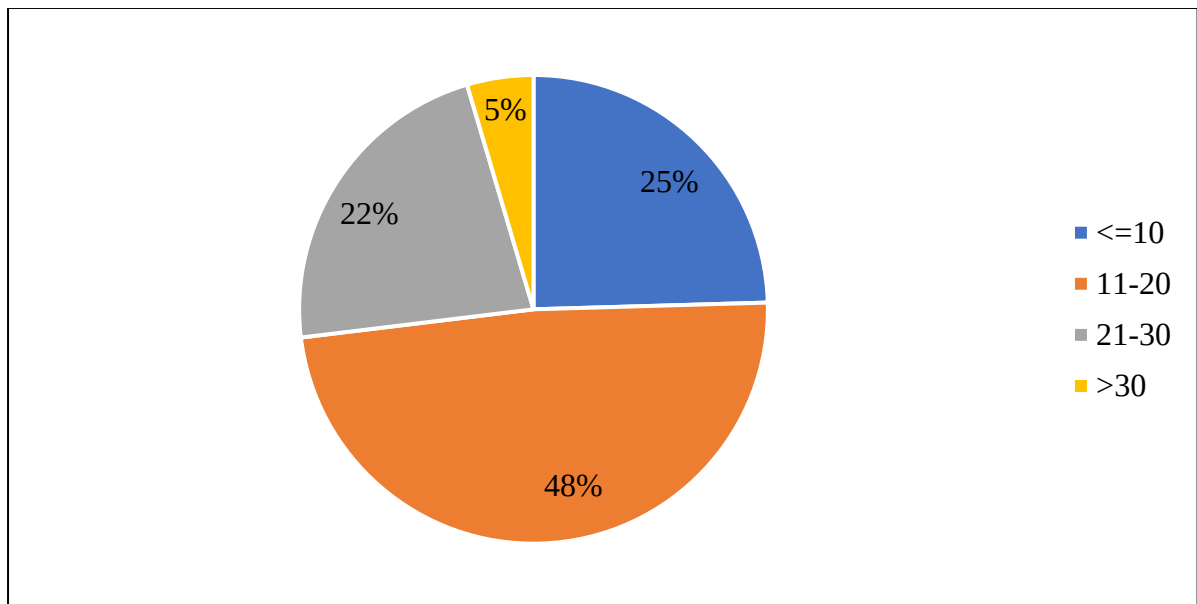
Source: Field Survey, 2022

Figure no. 5.13: Working Trip Time (Access-Egress-Direct)



Source: Field Survey, 2022

Figure no. 5.14: Non-Working Trip Time (Access-Egress-Direct)

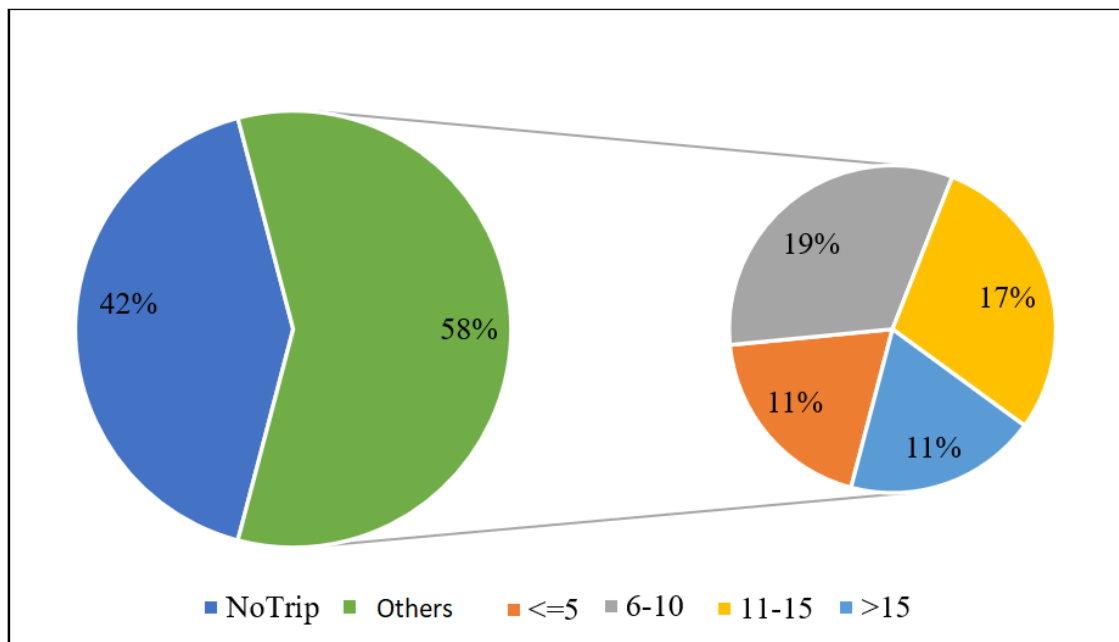


Source: Field Survey, 2022

Figure no. 5.15: Working Trip Time (Total)

Travel time of working and non-working trips are represented in figure 5.14 and 5.15. For working trip most of the worker travel between 11-20 min. a few travels between 21-30 min and less than 10 min. Workers travel greater than 30 min is very poor in number. As worker, prefer to reside near the working place for reducing their travel length. For non-

working trip almost 42% worker make no trip. Among rest 58% of the worker who conduct not working trip don't actually go very far from their house. They go to nearer bazar, shopping center for buying their daily basic commodities. But as most of the worker walk to their non-working trip's destination it increases their trip time a little bit. As a result, the trip time should also be low but not very which is clearly determined in the figure number 09. 19% of all non-working trip's time is 6 to 10 minutes. Second most percentage is found in the category of 11 to 15 minutes trip time.

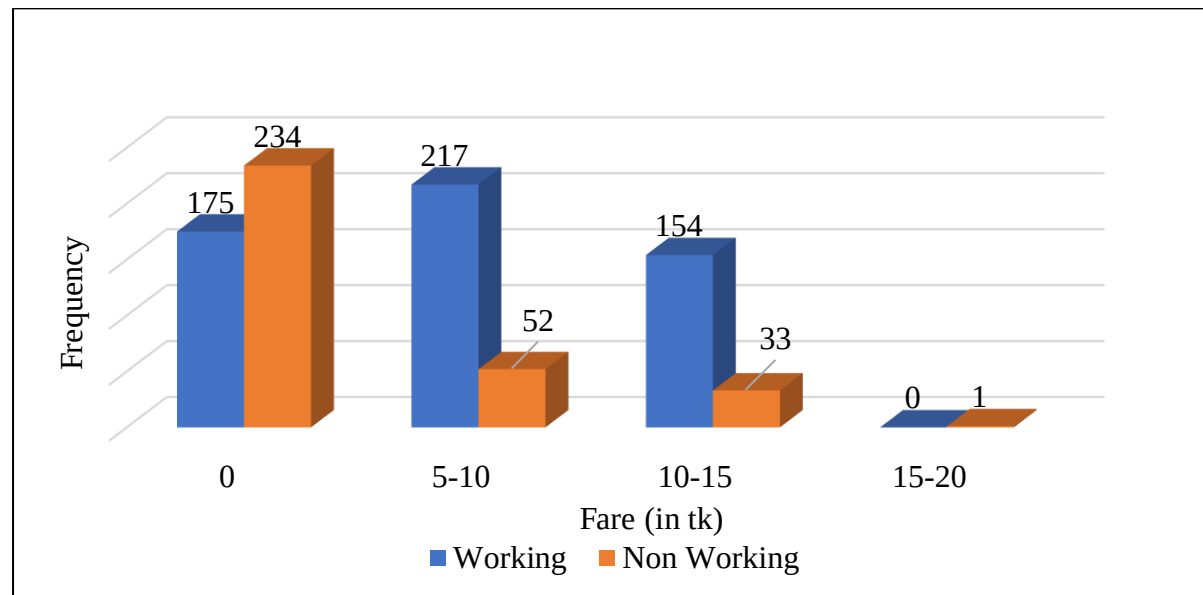


Source: Field Survey, 2022

Figure no. 5.16: Non-Working Trip Time (Total)

Besides, workers of other area also sometime get interested to visit for some recreation purpose which increase the number of frequencies of worker spending 11 to 15 minutes in their non-working trip time.

5.2.8 Fare

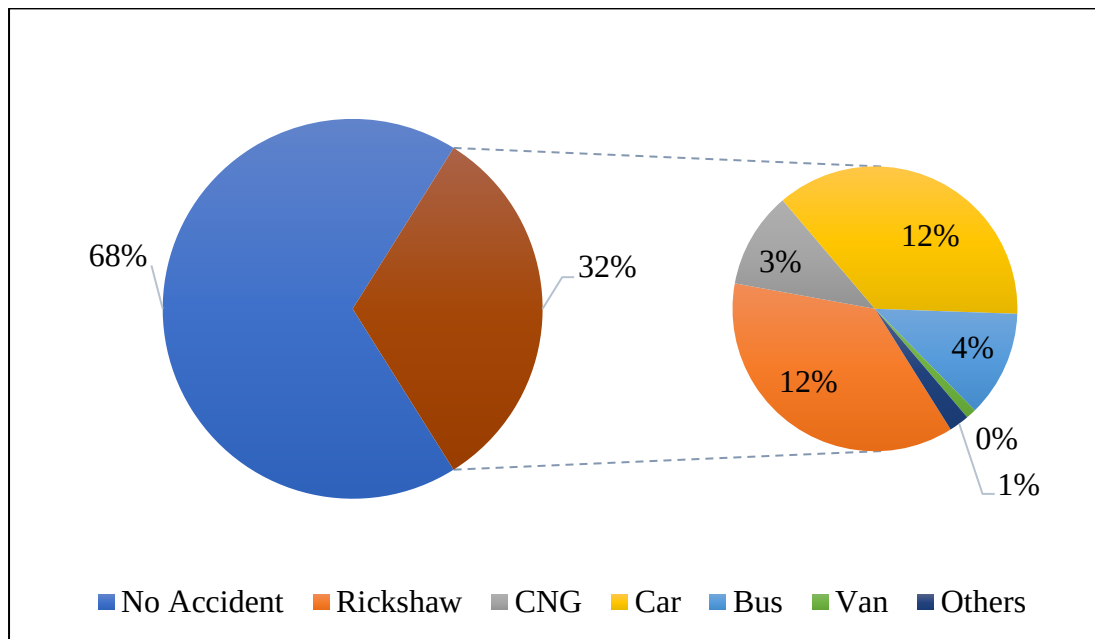


Source: Field Survey, 2022

Figure no. 5.17: Fare of the Trip (in taka)

Fare is a very important perspective that make worker decide to choose their mode. It is already found that the workers mainly walk to their destination for both working and non-working purpose, no fare will be needed for them. And as they do not travel much distance the fare shall also not be very high. It is been shown in the graph. Only 1 worker is found to spend more than 15 taka for non-working purpose. It is also found that workers are spending much in the working trip rather than non-working trip purpose.

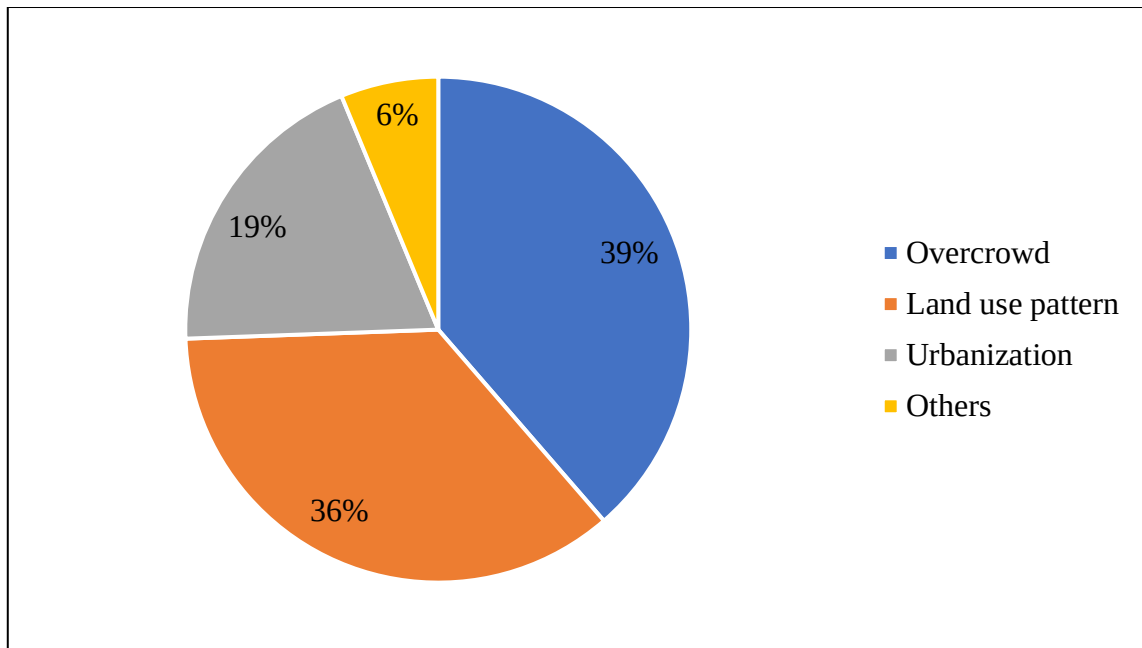
5.2.9 Accident During Trip



Source: Field Survey, 2022

Figure no. 5.18: Accident Conducting Vehicle

The rate of accidents conducting vehicles and causes of accidents in worker travel is presented in figure 5.18 & 5.19. The accident rate is 32% and most of the accident is happened by car and rickshaw. The main causes behind the vehicle accident are overcrowd and land use pattern. Also, urbanization is another influential cause in accidents. Due to rapid urbanization and population growth in the city unplanned & irregular land use pattern and overcrowdings in travel is increasing which are causing vehicle accidents.

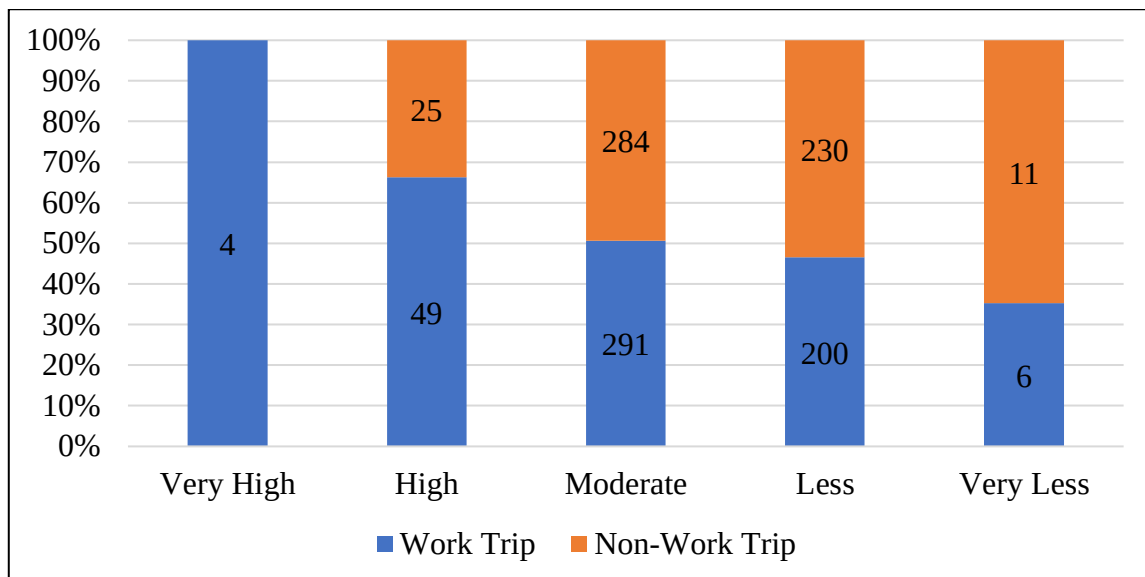


Source: Field Survey, 2022

Figure no. 5.19: Cause of Accident

5.3 Travel Satisfaction

5.3.1 Cost

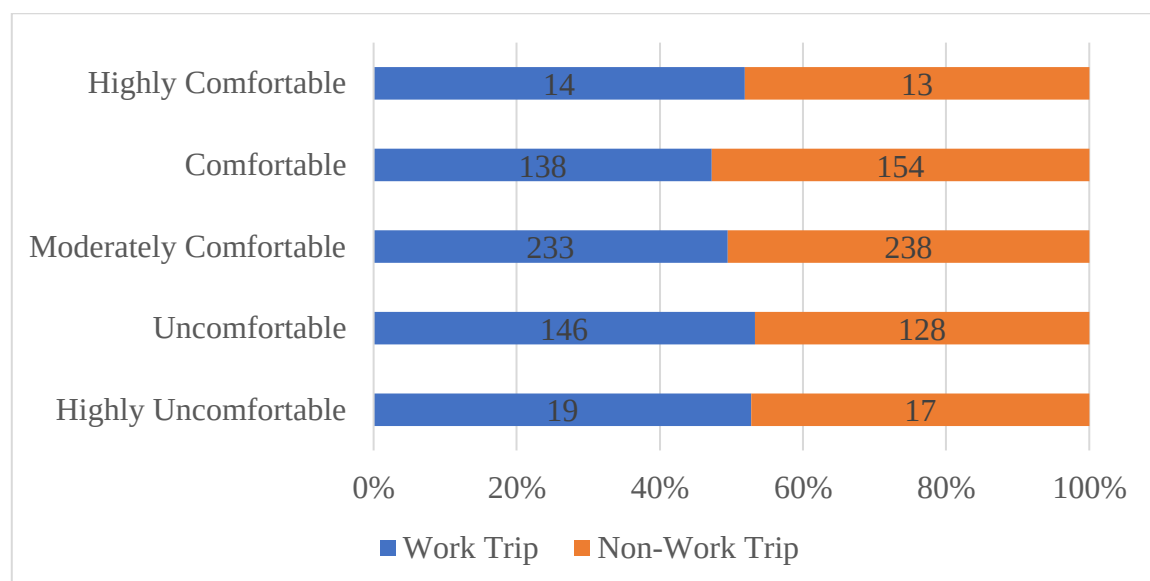


Source: Field Survey, 2022

Figure no. 5.20: Satisfaction on Cost

Cost is one of the most important satisfactory factors. This can affect the worker's mode choice highly. Figure 5.20 is showing the level of satisfaction of the workers on cost of their working and non-working trip. It can be seen that for satisfaction of cost for work trip is lower than non-work trip. The number of respondents saying the cost is very high to high is decreasing in non-work trip whereas this number is increasing for work trip. Around 50% of the workers have said that the cost on their trip both working and non-working is moderate. For non-work trip higher number of people has said the cost is very less than for work trip.

5.3.2 Comfort



Source: Field Survey, 2022

Figure no. 5.21: Satisfaction on Comfort

Satisfaction on Comfort of both work and non-work trip is been shown with a bar chart in figure no. 5.21. The figure is showing that among the worker who have said they are highly uncomfortable to their trip, 56% is during work trip and rest 44% is during non-work trip. Highly comfortability of both work trip and non-work trip is same. According to the workers they are more comfortable in their non-work trip rather than their work trip. As they mainly use walk as their main mode for non-work trip and their non-work trip's trip length is also not very high that's why comfortability of non-work trip is good. On the other hand, worker's mainly use bus and their trip length is also not as less as the lengths of non-

work trip. During peak hour, the condition of comfortability become much worse. That's why the comfortability of work trip length is not so good as of non-work trip.

5.3.3 Accessibility

Accessibility indicates mainly the distance to the mode. The percentage of different satisfactory level of accessibility is shown in figure 5.22 along with safety, security and reliability for work and non-work trip. The chart is showing that about 10% of the worker have commented that the accessibility is poor for working trip whereas 9% workers have commented so for non-work trip. Maximum percentage of workers have said that the accessibility of modes is fair to them in work trip but for in non-work trip it is good. 36% of workers have said their accessibility is good whereas 41% have said so for their non-work trip. It can be said that the overall accessibility is moderate.

5.3.4 Safety

The condition of safety of different mode is not so good as said by the workers. 23% of the worker have said the safety is poor for work trip. But the scenario is worse in non-work trip as 34% of workers have the safety condition is poor.



Source: Field Survey, 2022

Figure no. 5.22: Satisfaction on Accessibility, Safety, Security, Reliability

Though a good percentage have said their safety is fair but fair safety cannot be taken as a good sign as safety is a very sensitive factor which is highly responsible in taking decision on choosing mode.

5.3.5 Security

The scenario of the level of satisfaction on security of the workers is worse than safety both for working and non-working trip. 29% of worker have commented that the security of the modes is poor of work trip whereas 31% workers have said so about their non-work trip. Though 23% workers satisfaction on security is good but the overall satisfaction on security factors is not good.

5.3.6 Reliability

From the graph it can be easily understood that 68% of the workers are satisfied with the reliability of their work trip and among them around 10% have said it very good. On the

other hand, 65% % of the workers are satisfied with the reliability of their no-work trip and among them around 11% have said it very good.

5.3.7 Work Trip Modes vs Safety Crosstab

Table no. 5.7: Cross tabulation of Working modes vs Safety

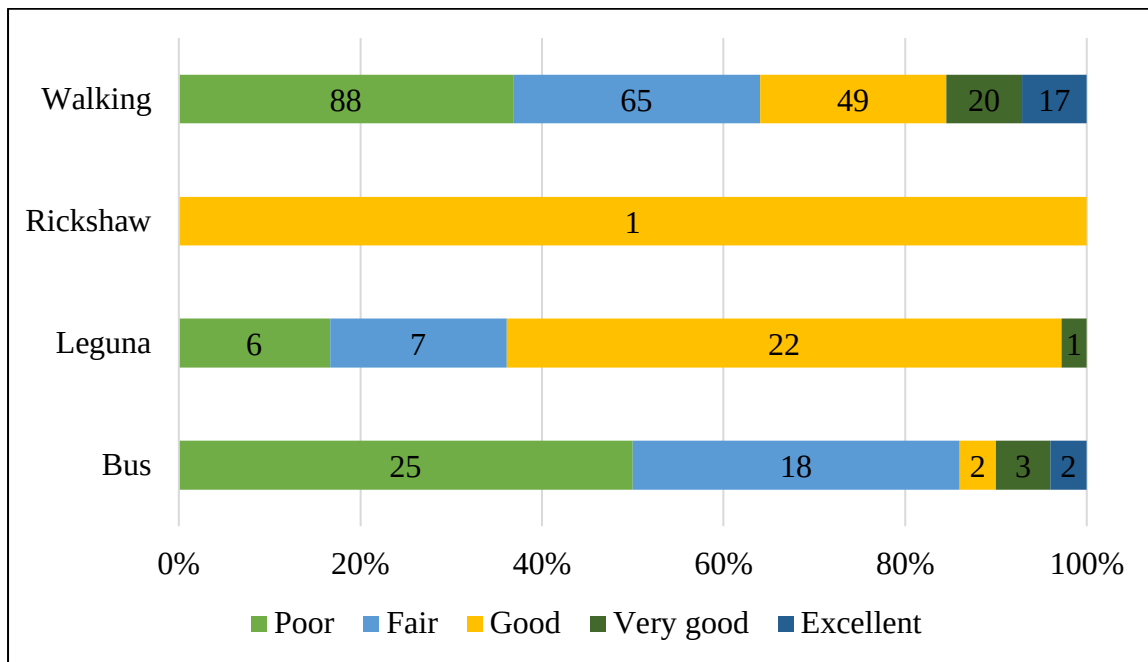
			Safety					Total
			Poor	Fair	Good	Very good	Excellent	
Mode	Bus	Count	59	75	27	5	0	166
		% within Mode	35.5%	45.2%	16.3%	3.0%	0.0%	100.0%
		% within Safety	46.8%	38.3%	18.5%	9.8%	0.0%	30.2%
		% of Total	10.7%	13.6%	4.9%	0.9%	0.0%	30.2%
	Leguna	Count	32	33	20	2	0	87
		% within Mode	36.8%	37.9%	23.0%	2.3%	0.0%	100.0%
		% within Safety	25.4%	16.8%	13.7%	3.9%	0.0%	15.8%
		% of Total	5.8%	6.0%	3.6%	0.4%	0.0%	15.8%
	Motorcycle	Count	1	1	0	0	0	2
		% within Mode	50.0%	50.0%	0.0%	0.0%	0.0%	100.0%
		% within Safety	0.8%	0.5%	0.0%	0.0%	0.0%	0.4%
		% of Total	0.2%	0.2%	0.0%	0.0%	0.0%	0.4%
	Walking	Count	27	81	57	6	0	171
		% within Mode	15.8%	47.4%	33.3%	3.5%	0.0%	100.0%
		% within Safety	21.4%	41.3%	39.0%	11.8%	0.0%	31.1%
		% of Total	4.9%	14.7%	10.4%	1.1%	0.0%	31.1%
	Bicycle	Count	0	0	2	0	0	2
		% within Mode	0.0%	0.0%	100.0%	0.0%	0.0%	100.0%
		% within Safety	0.0%	0.0%	1.4%	0.0%	0.0%	0.4%
		% of Total	0.0%	0.0%	0.4%	0.0%	0.0%	0.4%

	Others	Count	7	6	40	38	31	122
		% within Mode	5.7%	4.9%	32.8%	31.1%	25.4%	100.0%
		% within Safety	5.6%	3.1%	27.4%	74.5%	100.0%	22.2%
		% of Total	1.3%	1.1%	7.3%	6.9%	5.6%	22.2%
Total		Count	126	196	146	51	31	550
		% within Mode	22.9%	35.6%	26.5%	9.3%	5.6%	100.0%
		% within Safety	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
		% of Total	22.9%	35.6%	26.5%	9.3%	5.6%	100.0%

The relationship between working modes and safety on travelling those modes is given in table 5.7. In bus service majority of the workers believed that the safety provision of existing service is poor or fair. A very few respondents approved about good safety provision of bus service. The situation is also same for the leguna service provision. Most of the workers agreed that safety provision of leguna mode for working trip is not up to mark. It can be said that for working trip, bus and leguna modes safety provision is unsatisfactory. But walking as a mode of transport for working trip, according to the majority of their perspective safety provision good or neutral, though a few believed that walking mode safety is poor.

5.3.8 Relationship Between Non-Working Modes with Security of the Modes

Figure 5.23 is showing the satisfaction of security for different modes used in non-working trip. It has been said earlier that for non-working trip walk mode is mostly used by the worker. The security during walk is said poor by almost 40% of the worker. On the other hand, another 40% are satisfied with the security of walk in and among them around 10% have said that the security is excellent. But the scenario in the security of bus is worse than walk. Workers who use bus, around 50% of them have said the security is poor. Pickpocketing, harassment are the reasons that makes satisfaction of security that bad. A little amount of worker use leguna for their non-working trip and maximum of them have said that the security is good.



Source: Field Survey, 2022

Figure no. 5.23: Satisfaction on Accessibility, Safety, Security, Reliability

5.3.9 Work Trip Modes vs Reliability Crosstab

Table no. 5.8 this showing a crosstab of working mode with the reliability. It is clear from the analysis of the working mode that most of the passenger use bus as their working mode and walk after that. In the table it can be seen that among 171 of total bus user worker, 45.2% said the reliability is good. 4.8% of bus user have said the reliability is very good but 40.4% have described bus more not so good. Comparing to bus the reliability of walking mode is better as determined in the analysis. It is found that 72.5% of worker who walk to their working place have said this mode's reliability is good which is far better than bus mode. One of this mode users have also said the reliability of this mode is excellent. The best mode based on the reliability is institutional vehicle. 97.4% of total excellent reliability has been said by the user who use this mode. Besides, reliability of the users of leguna is also mostly good and then fair. Nearly half of the leguna user have described this mode as good. None of the user has said this mode's reliability excellent. From the total section it can be seen, 51.5% of the worker have said the mode's reliability is good, 10% have depicted this mode as very good and only 7% of user has voted for the excellent reliability

of modes. Fair reliability is said by 26% of the worker. From all the reliability value it can be summarized that the reliability is up to the mark. It is neither very good nor so much bad.

Table no. 5.8: Cross tabulation of Working modes vs Reliability

			Reliability					Total
			Poor	Fair	Good	Very good	Excellent	
Working Mode	Bus	Count	16	67	75	8	0	166
		% within working mode	9.6%	40.4%	45.2%	4.8%	0.0%	100.0%
		% within Reliability	55.2%	47.2%	26.7%	14.5%	0.0%	30.4%
		% of Total	2.9%	12.3%	13.7%	1.5%	0.0%	30.4%
	Leguna	Count	10	33	43	1	0	87
		% within working mode	11.5%	37.9%	49.4%	1.1%	0.0%	100.0%
		% within Reliability	34.5%	23.2%	15.3%	1.8%	0.0%	15.9%
		% of Total	1.8%	6.0%	7.9%	0.2%	0.0%	15.9%
	Walking	Count	3	36	124	7	1	171
		% within working mode	1.8%	21.1%	72.5%	4.1%	0.6%	100.0%
		% within Reliability	10.3%	25.4%	44.1%	12.7%	2.6%	31.3%
		% of Total	0.5%	6.6%	22.7%	1.3%	0.2%	31.3%
	Others	Count	0	6	39	39	38	122
		% within working mode	0.0%	4.9%	32.0%	32.0%	31.1%	100.0%
		% within Reliability	0.0%	4.2%	13.9%	70.9%	97.4%	22.3%
		% of Total	0.0%	1.1%	7.1%	7.1%	7.0%	22.3%
Total		Count	29	142	281	55	39	546
		% within working mode	5.3%	26.0%	51.5%	10.1%	7.1%	100.0%

	% within Reliability	100.0 %	100.0%	100.0%	100.0%	100.0%	100.0%
	% of Total	5.3%	26.0%	51.5%	10.1%	7.1%	100.0%

5.4 Regression Model for Work Trip

Table no. 5.9: Correlation

	Mod e	Age	Gend er	Inco me	Earnin g memb er	House hold numb er	Length trip	Trip Freque ncy	Acc ess Time	Egres s Time	Direc t Time	Fare
Mode	1	.040	.241	.476*	.103**	.120**	.325	-.112**	-.142**	-.134**	.423	-.514**
Age		1	-.310**	.207**	.105**	.259**	-.029	-.089*	-.186**	-.168**	.077*	-.182**
Gender			1	-.118**	.115**	-.003	-.144**	-.028	-.011	-.002	-.022	-.098*
Income				1	.341**	.487**	.211**	-.129**	.096*	.094*	.295**	.087*
Earning member					1	.386**	-.019	-.097*	-.001	-.026	.181**	-.089*
Household number						1	.133**	-.162**	-.030	-.073*	.304**	-.082*
Length trip							1	-.051	.324**	.264**	.369**	.203**
Trip Frequency								1	.046	.079*	-.098*	.149**
Access Time									1	.792**	.399**	.617**
Egress Time										1	.318**	.496**
Direct Time											1	.295**
Fare												1

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Correlation provides an overview about the relationship between the dependent and independent variables and the internal relation of independent variables also. The independent variables which are highly correlated or have high internal correlation with each other are not considered as suitable variable for analyzing regression model. For analyzing regression model, the independent variables which are highly correlated with the

dependent variable is selected and also have less internal correlation among them. In this paper the for working purpose trip gender, income of the household, length of the trip, direct trip time and fare of the mode are considered as independent variables or predictors for correlating with dependent variable-transportation mode choice and develop regression model. The correlation is also significant at .05 and .01 level which is defined by (*) and (**).

Table no. 5.10: Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.589 ^a	.346	.035	2.33648

a. Predictors: (Constant), fare, income, Gender, time direct, length.

Model Summary provides an overview of the results. From the table it has been seen that the value of R is 0.589 and the value of R^2 is 0.346 indicating that the explanatory variables entered into the model explain 34.6% of the variation of mode used for the work trip.

Table no. 5.11: Anova

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	133.590	5	26.718	4.894	.000 ^b
	Residual	2942.472	539	5.459		
	Total	3076.062	544			

a. Dependent Variable: mode

b. Predictors: (Constant), fare, income, Gender, time direct, length.

The above Table no 5.11 discuss about the test of significance of the regression model using an Anova. It shows whether the overall model is significant, $p = 0.000$. It also tells that if the overall result is highly significant, then at least 1 or more of the individual variables will most likely have a significant relationship to the dependent variable. It's mainly used in the identification of hypothesis for the prediction of result.

Table no. 5.12: Regression model for Work Trip

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	1.910	.567		3.370	.001
	Gender	.127	.209	.026	.604	.046
	Income	.295	.214	.062	1.375	.030
	Length	-.693	.288	-.134	-2.408	.016
	Time Direct	.583	.222	.123	2.630	.009
	Fare	.968	.278	.183	3.477	.001

The Regression Equation is

$$Y(\text{Mode}) = 1.910 + (0.127 \times \text{Gender}) + (0.295 \times \text{Income}) \\ - (0.693 \times \text{Length of the Trip}) + (0.583 \times \text{Direct Time}) + (0.968 \times \text{Fare})$$

In this regression model the independent variables directly and significantly affect the dependent variable. It is seen that only length is negatively related with the dependent variable and rest of the independent variables are positively related. Fare is very highly affecting the mode choice of the workers. Then the length is affecting the mode choice of the worker but negatively. There are some other factors like self-selection (e.g. attitudes, preference, behaviors etc.) and build-environment that influence the transportation mode choice badly but they are rarely addressed in this study. The regression model for non-working mode is validated as the pearson correlation value for 20% sample is higher than rest 80% sample.

5.5 Mode Choice Model for Work Trip

Here Bus and leguna, are used as reference to find the co-efficient of other mode choices. Below individual table are shown with description.

5.5.1 Bus as Reference

The coefficients and intercepts that were generated from the multinomial logit function and those that were found to be significant were used in the model. This model reveals that female workers mostly choose leguna and walk for their work trip.

Table no. 5.13: Logit Model Using Bus as Reference for Working Trip

Mode	Intercept	Gender	Income	Length of trip	Direct Trip Time	Fare
Leguna	-0.231	-0.779	-0.014	-0.056	0.571	-0.545
Walk	-20.277	0.434	0.266	2.885	-0.588	17.593

It is clear that bus is not suitable for women as the safety, security and comfort in bus is not so good. In the leguna female found more safety, security and comfort so as in walking. In respect of income factor, it is found from the model that worker whose household income are greater than 15 thousand are tend to choose leguna referring to bus. Workers are much concerned about their comfort, safety and security during their trip and this is the reason they choose these modes. On the other hand, worker whose income is less than 15 thousand usually prefer walking mode than bus as no cost is needed to walk to their working place. From the model it is clear that when the length is greater than 5KM workers are choosing leguna for their trip as comfort in these modes are better than bus. But when the distance of their trip from home to working place is less than 5KM they usually choose walk as it is not so hard to walk 1-2 KM than riding on bus. Time for the main trip which is direct time is found to be required lower in leguna and higher in walk referring to bus. As bus has many stoppages and it is obvious now a day that buses used to stop here and there for extra passenger, the required time in bus gets higher. In fare factor which the most influential factor than other it is found that comparing to bus the fare of leguna is 0.5 times higher. As this vehicle is providing more comfort, safety and security to the workers it is not so hard to believe that their fare would be higher so as in leguna. It is found in the walking mode that the fare is low. It is so true because in walk mode there is no fare actually. From the intercept value it can be said that workers are mostly tending to choose bus than other mode.

5.5.2 Leguna as Reference

Table no. 5.14: Logit Model Using Leguna as Reference Working Trip

Mode	Intercept	Gender	Income	Length of trip	Direct Trip Time	Fare
Bus	0.231	0.779	0.014	0.056	-0.571	0.545
Walk	-20.046	-0.344	0.280	2.941	0.017	18.139

The coefficients and intercepts that were generated from the multinomial logit function and those that were found to be significant were used in the model. This model reveals that, male workers are mostly choosing bus than leguna as the fare is lower in bus and males are not so concerned about the comfort and security of the modes. But in case of walk females are found to be choosing walk. Reason behind choosing these than mode than leguna is there is no cost of trip in walk and females are giving these factors more importance. Income is another very important factor which affect the mode choice of the workers. worker whose income is greater than 15thousand usually choosing institutional vehicle for their work trip. It us found that bus and walk mode is usually chosen by those workers whose household income is less than 15thousand. Reason behind this is the fare. In walk no cost is required and in bus the fare is lower than in leguna. In terms of length it is found from the model that workers are choosing bus and walk when their length of the trip is lower than or equal to 5KM. It is obvious that to travel shorter distance required time will also be short which is found in the trip time factor. The required time for work trip does not differ much in walk than in leguna. It is only 0.01times lower but in term of bus it is found that the required time is higher than leguna. In respect of fare people are choosing bus when it is less than 10tk. The intercept values are determining that workers are intended to choose bus than walk when leguna is taken as the reference in this model.

5.3 Regression Model for Non-Work Trip

Correlation provides an overview about the relationship between the dependent and independent variables and the internal relation of independent variables also. The independent variables which are highly correlated or have high internal correlation with each other are not considered as suitable variable for analyzing regression model. For analyzing regression model, the independent variables which are highly correlated with the dependent variable is selected and also have less internal correlation among them. In this

research for non-working purpose trip household income, length of the trip, direct trip time and fare of the mode are considered as independent variables or predictors for correlating with dependent variable-transportation mode choice and develop regression model.

Table no. 5.15: Correlation

	Mode	Age	Gender	Income	Earning member	Household number	Length of the trip	Frequency of the trip	Access Time	Egress Time	Direct Time	Fare
Mode	1	.235*	-.076	.281**	.159**	.234**	-.403**	-.233**	-.197**	-.263**	-.704**	-.764**
Age		1	-.316**	.442**	.179**	.414**	-.071	-.150**	-.178**	.056	-.231**	-.211**
Gender			1	-.351**	.040	-.178**	-.022	.076	.026	.005	.028	-.006
Income				1	.320**	.503**	-.037	-.106*	-.158**	-.032	-.087	-.147**
Earning member					1	.361**	-.074	-.109*	-.135**	-.055	-.100*	-.162**
Household number						1	-.104*	-.150**	-.211**	-.093*	-.131**	-.211**
Length of the trip							1	.198**	.369**	.076	.140**	.427**
Frequency of the trip								1	.171**	.034	.089	.108*
Access Time									1	.440**	.294**	.637**
Egress Time										1	-.008	.259**
Direct Time											1	.324**
Fare												1
*. Correlation is significant at the 0.05 level (2-tailed).												
**. Correlation is significant at the 0.01 level (2-tailed).												

Table no. 5.16: Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.787 ^a	.620	.615	1.000

a. Predictors: (Constant), Direct Time, Income, Length of the trip, Fare

Model Summary provides an overview of the results. From the table it has been seen that the value of R is 0.787 and the value of R^2 is 0.620 indicating that the explanatory variables entered into the model explain 62% of the variation of mode used for the work trip.

Table no. 5.17: Anova

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	512.765	4	128.191	128.279	.000 ^b
	Residual	314.785	315	.999		
	Total	827.550	319			

a. Dependent Variable: mode

b. Predictors: (Constant), Direct Time, Income, Length of the trip, Fare

The above discusses about the test of significance of the regression model using an Anova. It shows whether the overall model is significant, $p = 0.000$. It also tells that if the overall result is highly significant, then at least 1 or more of the individual variables will most likely have a significant relationship to the dependent variable. It's mainly used in the identification of hypothesis for the prediction of result.

Table no. 5.18: Regression model for Work Trip

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	4.860	.266		18.272	.000
	Income	.028	.010	.102	2.903	.004
	Length of the trip	-.320	.160	-.078	-1.999	.046
	Fare	-1.717	.099	-.707	-17.372	.000
	Direct Time	-.023	.014	-.060	-1.629	.042

a. Dependent Variable: mode

The Regression Equation is

$$Y = 4.680 + (0.028 \times \text{Household Income}) - (0.320 \times \text{Length of the Trip}) \\ - (1.717 \times \text{Fare}) - (0.023 \times \text{Direct Time})$$

Here Y = the modes used for non-working purpose.

In this regression model the independent variables directly and significantly affect the dependent variable. The maximum questionnaire survey was conducted from garment workers. From the equation it is seen that only household income is positively related with the dependent variable and rest of the independent variables are negatively related. Fare is very highly affecting the mode choice of the workers negatively. It means with the increase of fare workers are unwilling conduct non-working trip. Then the length is affecting the mode choice of the worker negatively. There are some other factors like self-selection (e.g. attitudes, preference, behaviors etc.) and build-environment that influence the transportation mode choice badly but they are rarely addressed in this study. The validation of this model has been shown in Appendix-B. The regression model for non-working mode is validated as the pearson correlation value for 20% sample is higher than rest 80% sample.

5.3.1 Mode Choice Model for Non-Work Trip

In assessing mode choice for non-working trip factors, which have significant influence are socio-economic factor such as income and travel factors such as length of trip, direct travel time and trip fare. For non-working trip it is seen from the figure 7.10 that most of the garment workers use bus, walk and leguna for their non-working trip. Using multinomial logit model, the coefficient of mode choice is determined taking bus and walk as reference.

5.3.2 Bus as Reference

In the first case of logit model bus is selected as reference, as 16% of garments workers use bus as a mode of transport for non-working trip.

Table no. 5.19: Logit Model Using Bus as Reference for Non-Working Trip

Mode	Intercept	Income	Length of trip	Direct Trip Time	Fare
Leguna	-.963	-.818	.580	0.803	-20.475
Walk	4.871	.782	3.680	-1.204	4.883

On the other hand, high income workers are taking leguna than bus. Length of the trip is one of the most important factors for the non-working trip. The trip value is recoded into two value where one is less than or equal to 5KM and another one is more than 5KM. It is

found that for greater than 5KM length of the trip workers are using bus rather than other mode. For very shorter distance they are actually not taking any motor vehicle. They simply walk there. Trip time is found less in leguna than bus as the length is lower. But for walking the trip time is higher as the coefficient is indicating negative value. Fare is much higher than bus for leguna and lower for walk. From the intercept value it can be seen that the value of walk is positive. So, it is clear that workers are more willing to take walk as their non-working mode.

5.3.3 Walk as Reference

In the second case of logit model walking is selected as reference, as 73% of workers use walking as a mode of transport.

Table no. 5.20: Logit Model Using Walk as Reference for Non-Working Trip

Mode	Intercept	Income	Length of trip	Direct Trip Time	Fare
Bus	-4.871	.782	-3.680	1.204	-2.007
Leguna	-3.909	-1.600	-3.100	2.007	-25.358

This table is showing the effect of some significant variable on bus and leguna for non-working mode comparing to walk. Workers having more than 15 thousand household income are actually taking leguna than walk and low-income workers are using bus for greater distance. Bus and leguna is mainly used for higher distance trip by the workers. Trip time is much less in leguna than bus comparing to walk. Fare is found to be higher than 5tk for both bus and leguna but it is found that the fare of leguna is much higher than bus.

5.3.4 Travel Satisfaction Level for Work Trip

Travel satisfaction is a very important term which mainly influence people to choose any mode and if the most important satisfactory factors can be developed people shall be choosier to use that type of mode. In this study several satisfactory factors are taken to demine the satisfaction level of workers. In this research it is found from the questionnaire survey that bus and walking are the mostly used mode of travel. The factors affecting choosing these vehicles are seen from the mode choice models. After the determination of mode choice, travel satisfaction level is determined to understand worker's intention and their importance on some specific factor. As bus is the highest used mode for work trip of the worker's, conjoint analysis is done on the bus user worker. The result is given in the following tables.

Table no. 5.21: Factor's Utility Value

		Utility Estimate	Std. Error
Cost	Very High	.096	.059
	High	.192	.071
	Moderate	.288	.116
	Less	.384	.139
	Very Less	.480	.168
Comfort	Highly Uncomfortable	.148	.085
	Uncomfortable	.296	.087
	Moderately Comfortable	.444	.096
	Comfortable	.592	.104
	Highly Comfortable	.740	.113
Accessibility	Poor	.051	.015
	Fair	.072	.021
	Good	.093	.056
	Very Good	.144	.042
	Excellent	.255	.077
Safety	Poor	.196	.185
	Fair	.392	.271
	Good	.588	.356
	Very Good	.784	.419
	Excellent	.980	.527
Security	Poor	.225	.285
	Fair	.330	.371
	Good	.555	.456
	Very Good	.670	.542
	Excellent	.825	.643
(Constant)		12.838	1.096

It can be found from the table that the standard deviation is less than the utility value which make the work much significant and it also means the error of the work is small also the data reflected actual situation.

The utility value reflects the preference about the respondent as the worker's subjective evaluation. It means that larger the utility value higher the preference. From table it can be seen that the utility value of cost is higher when it is very less and less in very low. In the comfort factor it is determined that worker's preference is higher in highly comfortability. Accessibility, safety, security's utility value is also higher for excellent service. It is normal that the preference of anyone will be for something which is the best. But which one they

are giving more importance is the most important thing that is to be determined for developing the specific satisfactory of the worker.

Table no. 5.22: Factor's Importance Value

Factor	Weight
Cost	19.191
Comfort	21.348
Accessibility	15.754
Safety	24.323
Security	22.383

Averaged Importance Score

According to the table the most important satisfaction determined is safety having a weight of 24.323. It is now a day a regular case that the accident of bus which take lots of life and it is increasing. So, the passengers are much more concern about the safety of their trip using bus mode. If the safety of the bus can be increased, it can result more attraction of user. Second important satisfactory factor is found security. From the research it is found that most of the workers are female. During the trip in bus they are being harassed and teased almost every day. The rate of eve teasing is increasing day by day which are one of the security issue of women. Besides snatching, stealing things inside bus has been a regular incident. That's why security factor is determined as the second most important factor with a weight of 22.383 which is not so less than safety. Third important factor which has been determined using conjoint analysis is comfort. Inside bus during office time it is seen that there is not any minimum space to even step the feet comfortably. This situation discourages the workers specially the female worker to use bus as their work trip mode. If the comfort level can be developed more, it will be able to encourage more worker. Cost and accessibility are the last two factors which the respondent have given less importance comparing to other factors and determined weight value is 19.19 and 15.75. The cost of bus is actually not so high in Chittagong and the number of bus is also good. That is the reason the workers are not so concerned about the cost and accessibility of bus at all.

Table no. 5.23: Correlation Value of Conjoint Analysis

	Value	Sig.
Pearson's R	.708	.020
Kendall's tau	.694	.028

a. Correlations between observed and estimated preferences

This table is mainly determining the significant of the model. Here Pearson's R and Kendall's tau value and significant value is shown. It can be seen from the table that both the values are near to 1 and the significant level is less than .05 which means that the results are very significant and the fitting accuracy is high. Therefore, it can be considered that the level's utility values are reasonable and reflect worker's preference.

5.3.5 Travel Satisfaction Level for Non-Work Trip

Previously for working trip travel satisfaction utility level, importance value is determined for bus as it is the mostly used mode by the worker. For non-work trip it is found that walk is the mostly used mode. That's why the travel satisfaction level is determined from the utility value and mainly form the importance value of different satisfactory factor for walk mode. Unlike the satisfactory factor for work trip cost is excluded in the conjoint analysis for non-work trip as in walk no cost is actually required.

Table no. 5.24: Factor's Utility Value

		Utility Estimate	Std. Error
Comfort	Highly Uncomfortable	.003	.161
	Uncomfortable	.006	.322
	Moderately Comfortable	.009	.483
	Comfortable	.012	.644
	Highly Comfortable	.015	.805
Accessibility	Poor	.031	.161
	Fair	.062	.322
	Good	.093	.483
	Very Good	.124	.644
	Excellent	.155	.805
Safety	Poor	.261	.161
	Fair	.522	.322
	Good	.783	.483
	Very Good	1.044	.644
	Excellent	1.305	.805
Security	Poor	.040	.161
	Fair	.080	.322
	Good	.120	.483
	Very Good	.160	.644
	Excellent	.200	.805
(Constant)		11.995	.992

It can be found from the table that the standard deviation is less than the utility value which make the work much significant and it also means the error of the work is small also the data reflected actual situation.

The utility value reflects the preference about the respondent as the worker's subjective evaluation. It means that larger the utility value higher the preference. From table it can be seen that the utility value of safety is higher when it is excellent and less in poor. It is same for accessibility and security. For comfort factor, the utility values are not so high which means the utility of comfort is not so high in walk mode. Among them which factor is more important can be found from the importance value in table no. 5.24

Table no. 5.25: Importance Value

Comfort	19.870
Accessibility	23.606
Safety	24.359
Security	32.166

Averaged Importance Score

From the importance value it can be seen that the importance value for security is much higher than any other factor. During walking there can be actually many security issues. Snatching, pickpocketing is very regular in now a day. The satisfaction level is mainly depending on the security during walking to any place for non-working purpose. Second most important factor is found the safety. This actually indicate the accident during walk by another vehicle. Because of many reasons like urbanization, mixed land use and so on, number of vehicles have been increased. The traffic rules are mostly not obeyed by the drivers. That's why number of accidents is increasing day by day. So, safety is a serious issue which need to be improved for increasing the satisfaction level if the garment workers. The least important value is fund for comfort. It indicates that the comfort is not so important satisfactory factor in walk mode for the workers. As they do not have to walk large distance for nonworking purpose this is not found as an important factor.

Table no. 5.26: Correlation Value of Conjoint Analysis

	Value	Sig.
Pearson's R	.647	.045
Kendall's tau	.198	.024

a. Correlations between observed and estimated preferences

This table is mainly determining the significant of the model. Here Pearson's R and Kendall's tau value and significant value is shown. It can be seen from the table that both the values are near to 1 and the significant level is less than .05 which means that the results are very significant and the fitting accuracy is high. Therefore, it can be considered that the level's utility values are reasonable and reflect worker's preference.

5.3.6 Overall Travel Satisfaction

From satisfaction value the overall travel satisfaction of both working and non-working trip is determined by calculating the overall satisfaction for each satisfactory factor and finally doing an average of all the satisfactory factor's value. The weightage for each scale has been given Satisfaction value is determined using the following equation.

$$SS = \frac{(n_1 \times w_1) + (n_2 \times w_2) + (n_3 \times w_3) + (n_4 \times w_4) + (n_5 \times w_5)}{N}$$

Here,

SS = Satisfaction Score

n_1 = Number of those respondent who selected response 1

w_1 = Weighting of response 1

n_2 = Number of those respondent who selected response 2

w_2 = Weighting of response 2

n_3 = Number of those respondent who selected response 3

w_3 = Weighting of response 3

n_4 = Number of those respondent who selected response 4

w_4 = Weighting of response 4

n_5 = Number of those respondent who selected response 5

w_5 = Weighting of response 5

N = Number of total respondents.

Weighting of different response is given in the appendix. In Likert chart for the satisfaction is arranged into 5 points. They are very dissatisfied, somewhat dissatisfied, fairly satisfied, very satisfied and completely satisfied (Peterson & Wilson, 1992). According to equation

used for calculating satisfaction value the highest value will be 5. So, based on the value of satisfaction the Likert chart can be use to define the satisfaction of the worker on both working and non-working trip.

From the table no. 5.27 it can be seen that, the satisfactory value for cost is highest comparing to other satisfactory factors with a value of 3.28 which indicates that the workers are very satisfied in respect of cost of the modes. On the other hand, the lowest value is found for safety factor with a value of 2.39 which indicate that workers are fairly satisfied of this sector. For non-working trip the scenario is same as the satisfaction of work trip. The overall satisfaction value is indicating that for both working and non-working trip workers are fairly satisfied.

Table no 5.27 Satisfactory Value

	Cost	Comfort	Accessibility	Safety	Security	Reliability	Overall
SV for Work Trip	3.28	2.97	2.63	2.39	2.41	2.88	2.76
SV for Non-Work Trip	3.41	2.99	2.75	2.06	2.23	2.81	2.70

Chapter -6

Findings & Recommendation

6.1 Major Findings

In the analysis chapter the socio-economic condition, travel behavior, travel satisfaction and the trip pattern is determined. Based of the analysis some major findings of these aspect are described below.

6.1.1 Aspect of Socio-Economic Condition

- In the industry of Agrabad male workers are engaged more than the female members.
- Most of the respondents age is between 23-27. This indicates that a good share of young people is engaged in this sector.
- The employed people have not high educational qualification. 62% of them have passed college.
- In 67% of household members are engaged in the different sectors. The main reason of this is the number of family size and low amount of salary per capita. In most of the household there are 4-6 member To maintain such household a good amount of money will be needed.
- 42% of the household's monthly income is in between 21 to 30 thousand taka. Only 6% household is earning more than 20 thousand BDT per month.
- Household income is highly proportional to the educational qualification. With the increase of educational qualification household income also increase. It is found that 88.9% of respondents who have passed only the primary level are earning a household income of 10 to 20 thousand taka per month. On the other hand, most of the college passed and graduate workers are getting more than 20 thousand taka.
- Most of the employed people travel to their working place in peer group which means they mostly do not travel alone but with some other persons who live beside or near to them.
- The major reason of choosing rented house is the accessibility of the working place.

6.1.2 Aspect of Travel Behavior

- The trip length of employed people is mainly between 1 to 5 KM for working & non-working purpose. As the respondents house is near to their working place they do not need to travel much. Besides, they do not tend to go to any distanced place for any recreational activity. Their main non-working trip is to go to the nearest shop, bazar, market for buying necessary commodities.
- 31% of workers use bus for their working trip. Bus is mainly used for longer distance. The trip time is also lower in bus & fare is also lower than other motorized vehicles.
- The mode choice changes with income level. People of high-income class is using Motorized vehicle which have no safety & security issue & the comfortability is good.
- The main reason of choosing bus is its cost effectiveness as said by the workers whereas leguna is said to be more time saving mode than any other which is the main reason of choosing this. Safety & security of this mode attracts them to use it.
- The total time spent in working trip by the worker is mostly between 11 to 20 minutes & only 5% of them require more than 30 minute to reach their working place. 42% of them do not usually conduct non-working trip & 30% of the rest spend up to 10 minute maximum in their trip.
- The fare of modes used by the employed person in work trips is not very high. The fare is between 5 to 10 taka. In non-wok trip, no fare is required for most of the passenger as they generally walk o their destination.
- 68% of workers have said they did not face any accident during both working & non-working trip. Those who have faced accident said that most of the accident is occurred by rickshaw & private car. But it does not cause any fatality or major injury. Major reason of accident is told overcrowding of vehicles & pedestrian during peak hour. The mixed land use pattern is also responsible.

6.1.3 Aspect of Travel Satisfaction

- Most of the workers have said that the cost of the modes they use is moderate. It indicates that the workers are moderately satisfied to the cost.
- Though most of the workers are indifferent about the comfort of the modes, maximum of rest workers have said that they are not comfortable to the modes they use.
- Overall satisfaction of security is found not good as only 40% of the workers are satisfied to the present security of the modes.
- Safety of the modes are also not so satisfactory.
- Among accessibility, safety, security and reliability employed people are mostly satisfied about the reliability of the modes.
- The overall safety on bus and leguna is not so satisfactory to the respondents. Walk is considered as the safest mode.

6.1.4 Mode Choice Model

- For the mode choice of work trip, the most significant variables that affect the mode choice are gender of the worker, household income, length of the trip, direct trip and fare.
- It is found that female prefer leguna & walk as their mode choice.
- Leguna is more preferred by those respondents whose income is more than 20 thousand taka per month. Workers mostly prefer walk than bus whose monthly income is less than 20 thousand taka per month.
- When length of the trip is less than 5 km, employed persons mostly walk to their destinations. For longer distance they prefer leguna than bus.
- Direct trip time in walking required more than any other vehicle. The trip time is found to be the lowest in leguna.
- The intercept value of both models indicate that bus is the most preferred mode for work trip.
- In non-work trip the significant variables mainly affecting the mode choice of worker are household income, length of the trip, fare & direct trip time. Only gender is not significant.

- For the model bus & walk is taken as reference. People in non-work trip prefer walking whose income is less than 20 thousand taka.
- For less trip length, walk is mostly preferable by the respondents.
- Direct trip time is higher in walking comparing to bus & leguna.
- Fare is very much higher for leguna than bus as in walk no fare is required.
- From both of the model, it is found out that walking is the most preferred mode for the employed people in non-work trip.

6.1.5 Travel Satisfaction Level

- From the utility value determined by conjoint analysis it is found that the utility of the worker is getting higher as it moves from poor to excellent for accessibility, safety, security, reliability, very high to very less for cost and highly uncomfortable to highly comfortable.
- The most important satisfactory factor for worker in work by bus trip is the safety of the mode with an importance value of 24.323 determined by conjoint analysis. Second most important factor is security. Accessibility has the least importance factor which indicates that accessibility doesn't really affect the satisfaction level.
- Security is found to be the most important satisfactory factor with a value of 32.166. Second most is safety. These two are the most important satisfactory factor which actually determine the satisfaction of non-work trip of the workers.
- The satisfactory score of cost of work trip mode and non-work trip mode is 3.28 and 3.41 respectively which indicates that workers are very satisfied with both of the modes.
- In work trip modes and non-work trip modes employed people are the least satisfied with the safety as the satisfaction score of its the lowest for both among other factor's scores.
- In overall it can be said that the respondents are fairly satisfied with both work trip and non-work trip modes.

6.2 Recommendation

- As it is found earlier that the institutional vehicles are better for both reliability and safety. So, it is recommended that the number of the institutional vehicles should be increased as much as possible.
- The housing development should focus on the workers keeping in mind their income level and socio-economic conditions.
- As bus is used heavily by the workers, the number of the buses should be increased as well as the safety and security of the buses.
- Vehicular safety should be ensured especially for the women.
- Any types of harassment should be prohibited specially for the female workers.
- Rescheduling of the working hours is important to avoid congestion in the roads.
- For the drivers and the staffs of the transport services, awareness program is needed to organize in which the proper traffic rules and regulations will be taught to ensure reduced accidents or other.
- Provision of better safety and convenient environment inside the industries and also outside of working place.
- Female friendly transport service for example bus service dedicated only for female should be introduced.

Chapter -7

Conclusion

For a developing country like Bangladesh public & private sector plays a very vital role in economic development especially in generating employment as well as in foreign exchange earnings. Chittagong which is known as commercial capital of Bangladesh & most of the multinational firms & banks are situated in this area. So, these employed people are making different working and non-working trips using different mode of transport in their daily life. The study is undertaken to analyze the mode choice factor of employed people and determine the satisfaction factor and satisfaction level of their travel. However, this study does not consider the self-selection and built-environment factors. As the study only assessing the mode choice of only a few numbers of employed people, farther studies can be extended on mode choice of all employed people of the city. The study applied the multinomial logistic model for analyzing mode choice, as there is more than two category of mode and for determining the factors affecting mode choice of employed people, working and non-working trip multilinear regression analysis is adopted. Conjoint analysis is applied for identification of significant travel satisfaction factor and their utility level and a weighted index model is applied for determining travel satisfaction level. The results showed that the trip length of the workers is mainly between 1 to 5 km for both working and non-working purpose as majority of the their house is rented near to their working place and most of them chose bus for working trip and walking for non-working trip as travel mode. With the increase of income mode choice also change and with high income they are more inclined to use motorized mode of transport for better safety and comfortability. For the mode choice of work trip, the most significant variables that affect the mode choice are gender of the worker, household income, length of the trip, direct trip time and fare. Female workers mostly prefer leguna and walk. In non-work trip, the significant variables are same as the work trip factors except gender, which is not significant for non-working trip. For the model bus and walk is taken as reference for both work trip and non-work trip. People's whose income is less than 20 thousand taka prefer walk. In travel satisfaction of the workers among accessibility, safety, security and

reliability respondents are mostly satisfied about the reliability of the modes. The workers are moderately satisfied to the cost as according to them the cost of the mode they use is moderate. The overall safety on bus and leguna is not so satisfactory to the people and they considered walk as the safest mode. According to the respondents intuitional or official vehicle is said to be the best reliable mode than any other for work trip. The satisfactory score of cost of work trip mode and non-work trip mode is 3.28 and 3.41 respectively, which indicates that workers are very satisfied with both of the modes. From the utility value determined for identifying significant factor of travel satisfaction, it is found that the utility of the worker is getting higher as it moves from poor to excellent for accessibility, safety, security, reliability, very high to very less for cost and highly uncomfortable to highly comfortable. And according to majority of the workers for bus safety and security is the most significant factor and accessibility is the least significant factor. This study is providing a comprehensive outlook of the factor affecting the mode choice and important satisfactory factor among comfortability, reliability, safety, security, accessibility that will be useful for further research on employed people. The findings can be used to examine the traffic jam because of the trip in further research and developing the road network system. Also, in developing the land use pattern, in future, which is needed for balancing the mixed land use the study will be useful.

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